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by

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A Knowledge Management Approach to Rainfall Data: Design and Implementation of a Centralized System for the Tunisian Water sector

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Dedication

I dedicate this work to the soul of my father, whose presence continues to guide and inspire me every day. To my mother, for her unconditional love, sacrifices, and unwavering support, without which I could never have reached this level.

To my grandmothers, for their tenderness, prayers, and constant encouragement that have always illuminated my path.

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List of abbreviations

AMC	Conceptual Analysis and Modeling (Analyse et Modélisation Conceptuelle)
API	Application Programming Interface
BI	Business Intelligence
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
CRDA	Regional Agricultural Development Commissariat
DGACTA	General Directorate of Land Development and Agricultural Conservation
DGRE	General Directorate of Water Resources
ER	Entity-Relationship
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IT	Information Technology
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
KM	Knowledge Management
KMC	Knowledge Management Cycle
KMS	Knowledge Management System
MARHP	Ministry of Agriculture, Water Resources and Fisheries
MCD	Conceptual Data Model (Modèle Conceptuel de Données)
MCT	Conceptual Process Model (Modèle Conceptuel de Traitements)
MERISE	Method for Information Systems Analysis and Design
NGO	Non-Governmental Organization
ONAGRI	National Observatory of Agriculture
OWASP	Open Worldwide Application Security Project
PDF	Portable Document Format
PESTEL	Political, Economic, Sociocultural, Technological, Environmental, and Legal
PLUVIOM	Pluviometric Management System
REST	Representational State Transfer
SQL	Structured Query Language
SWOT	Strengths, Weaknesses, Opportunities, Threats
UI	User Interface
UX	User Experience

Abstract

This thesis addresses the critical challenge of rainfall data management in Tunisia, a country facing increasing water scarcity. The current system, which is fragmented and inconsistent, relies on manual processes that lead to errors and inefficiencies. In response, this project proposes the modernization of the rainfall data management system through the creation of a centralized web application.

Based on a methodology combining Design Thinking and the Lead User approach, the project began by thoroughly analyzing user needs and the weaknesses of the existing system. A strategic analysis using SWOT and PESTEL tools confirmed the relevance and feasibility of the proposed solution.

The core of this work involved designing and developing a robust postgres database (PLUVIOM) using the MERISE method, and creating a web application that allows for real-time data entry, consultation, cartographic visualization, and management of rainfall data. This solution aims to replace manual processes, ensure data consistency, and provide centralized, reliable access to crucial information for water resource management in Tunisia.

Résumé

Ce mémoire aborde le défi critique de la gestion des données pluviométriques en Tunisie, un pays confronté à une pénurie d'eau croissante. Le système actuel repose sur des processus manuels qui entraînent des erreurs et des inefficacités.

En réponse à ces défis ce travail propose la modernisation du système de gestion des données pluviométriques à travers la création d'une application web centralisée.

Basé sur une méthodologie combinant le Design Thinking et l'approche Lead User ce travail a débuté par une analyse approfondie des besoins des utilisateurs et des faiblesses du système actuel. Une analyse stratégique à l'aide des outils SWOT et PESTEL a confirmé la pertinence et la faisabilité de la solution proposée. Le travail a consisté à concevoir et développer une base de données PostgreSQL nommée PLUVIOM) en utilisant la méthode de design MERISE, et à la création d'une application web pour la saisie, la consultation, la visualisation cartographique et la gestion en temps réel des données pluviométriques. Cette solution vise à remplacer les processus manuels pour assurer la cohérence des données et fournir un accès centralisé et fiable à des informations pour la gestion des ressources en eau en Tunisie.

ملخص

تتناول هذه الأطروحة تحدي إدارة بيانات هطول الأمطار في تونس. يعتمد النظام الحالي على عمليات يدوية تؤدي إلى الأخطاء وعدم الكفاءة. واستجابة لذلك، يقترح هذا المشروع تحديث نظام إدارة بيانات هطول الأمطار من خلال إنشاء تطبيق ويب مركزي.

استنادًا إلى منهجية تجمع بين التفكير التصميمي (Design Thinking) ونهج المستخدم الرائد (Lead User)، بدأ المشروع بتحليل شامل لاحتياجات المستخدم ونقاط ضعف النظام الحالي. وأكد التحليل الاستراتيجي باستخدام أدوات SWOT و PESTEL أهمية الحل المقترح وجدواه.

تمثل جوهر هذا العمل في تصميم وتطوير قاعدة بيانات PostgreSQL (PLUVIOM) باستخدام طريقة MERISE، وإنشاء تطبيق ويب يتيح إدخال البيانات والإطلاع عليها وتصورها على الخرائط وإدارتها في الوقت الفعلي. يهدف هذا الحل إلى استبدال العمليات اليدوية، وضمان اتساق البيانات، وتوفير وصول مركزي وموثوق إلى المعلومات الحيوية

Introduction Générale

I. Contextual Background: Water Scarcity and National Development

Water plays a central role in Tunisia's development, yet managing this resource effectively remains a persistent challenge. The Ministry of Agriculture, Water Resources, and Fisheries (MARHP) oversees both agricultural production and water resource management a combination that places significant pressure on a single institution. Agriculture alone uses nearly 80% of the country's available water resources (Water Sector Report, 2023).

Balancing the growing water demands of agriculture with the need to preserve water for other uses, such as domestic supply and environmental conservation, creates a complex situation. To navigate this, the Ministry has adopted new approaches like social, open, service, and eco-innovation. These strategies aim to modernize the water sector and promote sustainable resource use (MARHP Website, 2025).

Despite these efforts, one of the major obstacles that still hampers progress is the ineffective management and exchange of information, particularly when it comes to hydrological data.

II. Problem Statement: Data Fragmentation and Inconsistency

A key challenge in the water sector today is the fragmented management of data. In many cases, information is not openly shared between departments or institutions. This situation is often the result of working habits where each structure sees the data it collects as its own, limiting cooperation and preventing broader access to information.

This lack of communication can lead to duplication of effort, where different teams unknowingly work on similar tasks or repeat work that has already been done. These issues are especially visible in the handling of rainfall data, which plays a critical role in national water management.

Currently, a daily rainfall update is posted on the ONAGRI website. This involves collecting data from the 24 regional agricultural offices (CRDAs) each morning, entering it into an Excel file, and publishing it on the same day. However, when comparing these daily figures with those stored in the annual Access database, differences in recorded values are often found. This situation raises an important question: Which dataset is more trustworthy—the real-time daily report, or the yearly aggregated records?

This discrepancy highlights the absence of a consistent process for verifying and consolidating rainfall data. Since this information is used by other departments within the Ministry, as well as by researchers and universities, improving data reliability is essential.

The goal of this project is to enhance the management of rainfall data from its collection to its publication. By streamlining the process and reinforcing communication between regional and central units, the project seeks to ensure that rainfall records are both accurate and easily accessible.

The solution involves developing a more coordinated and transparent data system that reduces errors, improves efficiency, and supports better decision-making. This initiative contributes to building a more innovative, collaborative, and sustainable water management system in Tunisia.

Chapter I: The General Directorate for Water Resources (DGRE)

I. Introduction

This first chapter provides a detailed analysis of The General Directorate for Water Resources (DGRE), is a National Directorate under the Ministry of Agriculture, Water Resources, and Fisheries (MARHP), specialised in a water resources in general covering the measurement, the monitoring, the preservation, and reporting of the nation's water resources.

a detailed analysis of the DGRE's organizational structure, its key missions, the existing data landscape it oversees, and the specific hydrological context of Tunisia, which highlights the need for an integrated knowledge management system.

II. Missions and Responsibilities of the DGRE

The DGRE's operational scope is defined by the decree of February 13, 2001 (Jort, 2001), which outlines its primary responsibilities across three strategic areas: monitoring, policy implementation, and innovation.

II.1 Monitoring and Evaluation of Water Resources

The foundational mission of the DGRE is to maintain a comprehensive and accurate understanding of the country's water status. This involves:

- ✓ **Establishing and Managing Measurement Networks:** The DGRE operates and maintains extensive networks for hydrology (river flow, dam levels), meteorology (rainfall, temperature), and piezometry (groundwater levels). The integrity and density of these networks are paramount for reliable data collection.
- ✓ **Data Collection and Analysis:** The institution is responsible for the systematic collection, processing, and analysis of hydrological data to ensure effective resource monitoring. This process often involves complex data validation and quality control procedures.
- ✓ **Preparing Water Balance Reports:** The DGRE regularly produces national and regional water balance reports, which are essential for strategic planning, resource allocation, and assessing the impact of water policies.

II.2 Development and Implementation of Water Management Policies

Beyond monitoring, the DGRE is the primary driver of national water policy, focusing on sustainability and efficiency:

- ✓ **Defining Exploitation Strategies:** It develops long-term strategies for the sustainable exploitation of both conventional (surface and groundwater) and non-conventional water resources (treated wastewater, desalinated water).

- ✓ **Developing Management Plans:** This includes creating detailed management plans for specific water basins and regions, ensuring integrated water resource management (IWRM) principles are applied.
- ✓ **Implementing Recharge Programs:** The DGRE is actively involved in implementing programs for the artificial recharge of groundwater aquifers to combat over-exploitation and rising salinity.

II.3 Promotion of Research and Innovation

Recognizing the challenges posed by climate change, the DGRE promotes technological integration and research:

- ✓ **Integrating New Technologies:** This involves the adoption of advanced tools such as remote sensing, Geographic Information Systems (GIS), and artificial intelligence for predictive modeling and operational management.
- ✓ **Participating in Research Projects:** The DGRE collaborates in national and international research projects focused on water resources management, aiming to incorporate the latest scientific findings into its operational practices.

III. Organizational Structure and Departmental Functions

The DGRE is structured into specialized departments to efficiently cover all facets of water management (Jort, 2001).

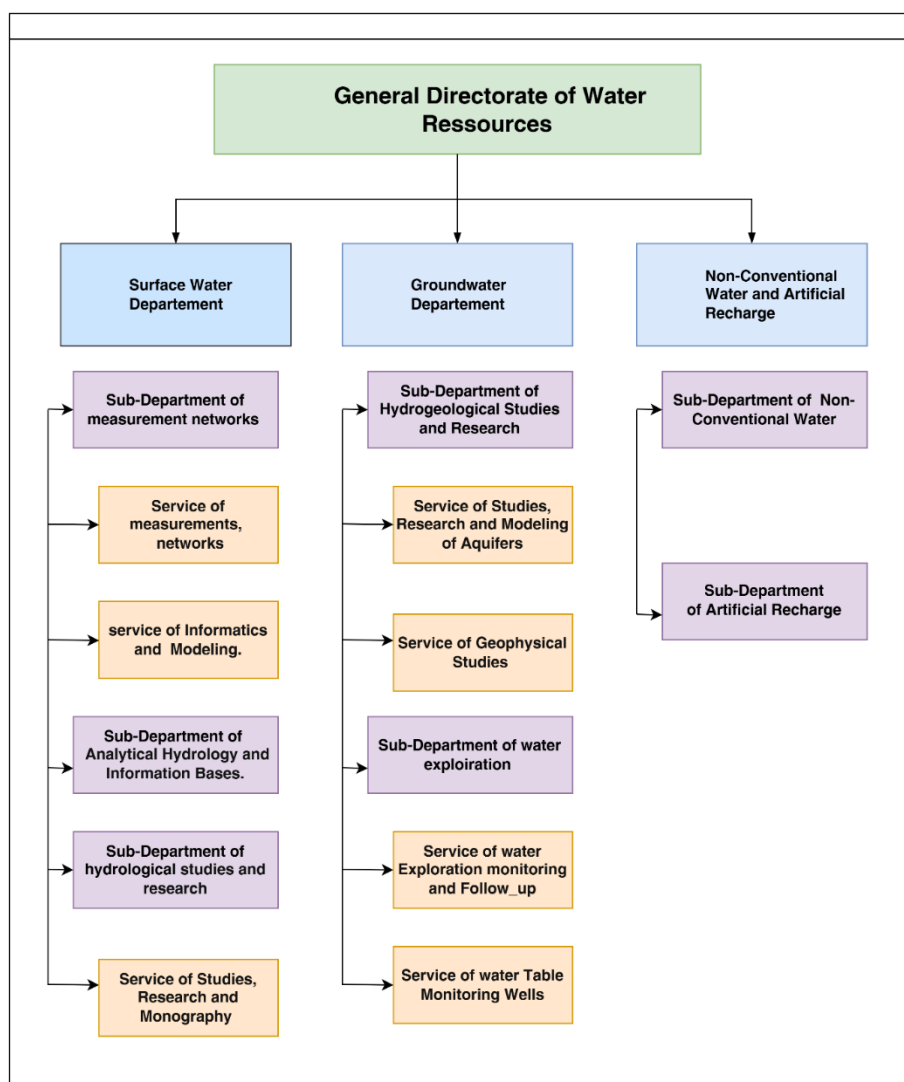


Figure 1 : Organizational Chart of the DGRE

The three core water resource management departments are central to the DGRE's operations:

III.1 Surface Water Department

This department focuses on the visible components of the hydrological cycle:

- **Network Management:** Establishing and managing measurement and monitoring networks for surface water bodies (rivers, dams).
- **Flood Management:** Developing early warning and flood declaration systems, a critical function for public safety and infrastructure protection.
- **Data Analysis and Modeling:** Applying modern technologies for data analysis, developing hydrological databases, and conducting studies to assess surface water resources. This includes the development of mathematical modeling techniques for predicting flow and storage.

- **Budgeting:** Setting hydrological budgets at the national and regional scale to inform allocation decisions.

III.2 Groundwater Department

Given Tunisia's reliance on aquifers, this department is crucial for managing hidden resources:

- **Hydrogeological Studies:** Preparing geophysical and hydrogeological studies to accurately assess groundwater resources and their sustainability.
- **Exploitation Programming:** Programming groundwater exploitation based on supply and demand forecasts, often involving complex regulatory measures.
- **Monitoring Networks:** Developing and maintaining piezometric networks to monitor groundwater reserves, water levels, and quality changes.
- **Modeling and Research:** Applying mathematical modeling techniques to adjust the water balances of underground reservoirs and conducting specialized research in hydrogeology.

III.3 Non-Conventional Water Department

This department addresses the strategic imperative of diversifying water sources:

- **Resource Assessment:** Quantitatively and qualitatively assessing non-conventional water resources, primarily treated wastewater and brackish/saline water.
- **Exploitation Development:** Carrying out studies and projects to develop the exploitation and use of these resources, ensuring compliance with environmental standards.
- **Artificial Recharge:** Programming and monitoring artificial recharge activities for underground reservoirs, including studying the effects of this feeding on water table reserves.

IV.3 Data Utilization and Strategic Importance

The DGRE centralizes a massive and diverse set of data collected at various frequencies (daily, monthly, yearly) and includes a wide range of parameters. The data managed by the DGRE is used for a multitude of high-stakes purposes, underscoring the necessity of its reliability:

- **Operational Reporting:** Preparation of daily, monthly, and annual hydrological bulletins and reports.
- **Infrastructure Design:** Providing essential inputs for the design and safety assessment of hydraulic structures (reservoirs, dams, flood protection).

- **Scientific Research:** Serving as the primary source for technical papers and research on Tunisian hydrology.
- **Decision Support:** Informing critical decisions regarding irrigation scheduling, water allocation, and flood risk management.

The figure 2 illustrates the various annual directories and reports published by the DGRE, such as daily rainfall reports, annual hydrological reports, and water quality reports, emphasizing the output of the data management process

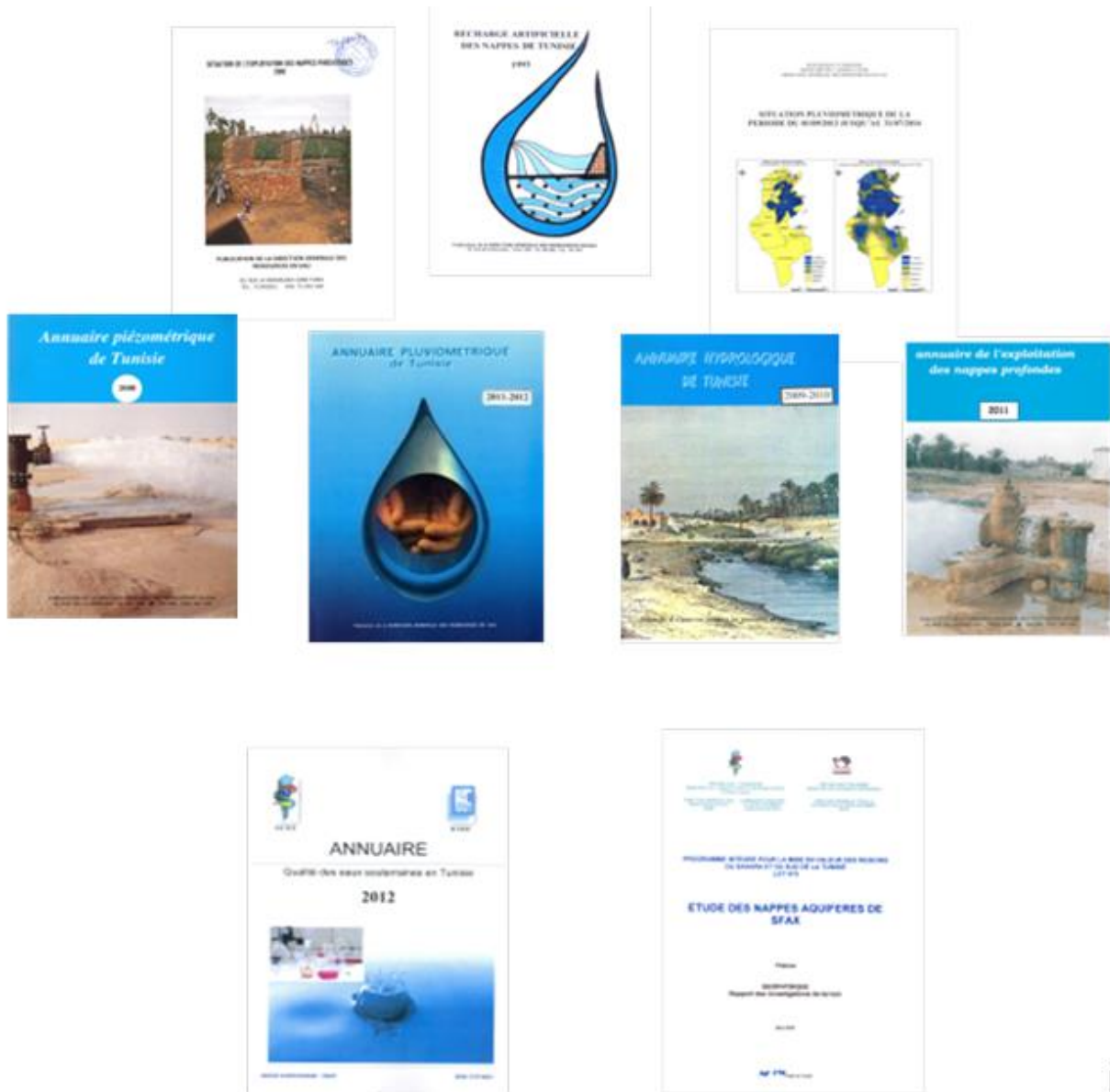


Figure 2 : The DGRE Publications

IV. CONCLUSION:

Chapter I has established the DGRE as a vital institution operating in a context of severe water stress. Its missions are complex, demanding high levels of data accuracy and integration. The analysis of the existing data landscape reveals a critical vulnerability.

This fragmentation hinders the transformation of raw data into reliable, actionable knowledge, thereby compromising the DGRE's ability to execute its strategic mandate effectively. The transition from a data-centric to a knowledge-centric approach, guided by modern KM principles, is essential to overcome these operational limitations. The following chapter will explore the theoretical frameworks necessary to design this integrated solution.

Chapter II: Literature review

I. Introduction :

This chapter presents the main theoretical concepts used in this report, focusing on knowledge management and its role in improving organizational performance. It explains how raw data is transformed into useful information and then into actionable knowledge to support decision-making, innovation, and coordination among stakeholders.

The chapter also introduces Design Thinking, a human-centered, creative approach that helps develop practical solutions through empathy, ideation, and implementation. SWOT and PESTEL analyses are presented as tools to evaluate internal strengths and weaknesses, as well as external opportunities and risks. Finally, the role of IT technologies in knowledge management is highlighted, showing how digital tools, knowledge systems, and collaboration platforms help capture, store, share, and use knowledge efficiently.

II. Definition of Knowledge

Data, information, and knowledge represent three distinct levels of processing within an organization (Davis & Botkin, 1994)

- **Data:** These are raw and objective facts that have no meaning by themselves. They form the basic material for any knowledge management process.

- **Information:** This is data that has been related and interpreted to give it meaning. Information can be useful or not, depending on the context in which it is used.

- **Knowledge:** This comes from combining information with understanding and reasoning. Knowledge is a structured process that helps use information effectively to solve problems and make informed decisions.

In short, efficient knowledge management involves converting raw data into useful information and then transforming that information into actionable knowledge, which in turn drives innovation and improves organizational performance.

Knowledge is more than just data or information, since data are simply isolated facts without interpretation, and there is also a need to clearly distinguish between information and knowledge in today's management practices (Davenport & Prusak, 1998)

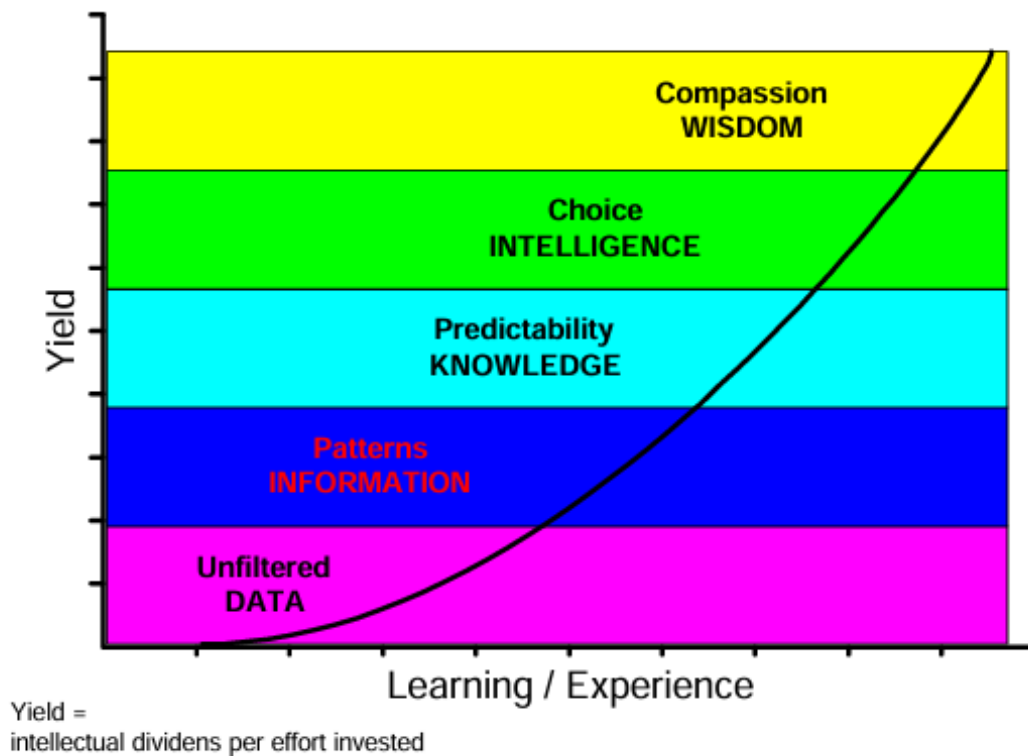


Figure 3 : Knowledge hierarchy , (Tuomi, 1999)

Moreover, some researchers challenge the traditional hierarchy by proposing that knowledge actually comes first, followed by information and then data, highlighting the crucial role of both mental and social processes in forming organizational knowledge (Tuomi, 1999). For our study, we have chosen the approach best suited for the Tunisian water sector, focusing on transforming raw data into meaningful and practical information

III. Types of Knowledge

In knowledge management, knowledge is typically classified into two main categories (Nonaka & Takeuchi, 1995)

Explicit Knowledge:

this type of knowledge is well-documented and codified. It appears in written formats such as reports, manuals, procedures, and databases, which makes it easy to share, store, and transfer.

Tacit Knowledge:

Tacit knowledge is personal and informal. It is based on individual experiences, skills, intuitions, and values. Because it is often unspoken and deeply ingrained through experience and social interactions, tacit knowledge is much harder to express or transfer.

IV. Knowledge Management

Knowledge management is a formal process of creating, sharing, and using knowledge within an organization to improve its efficiency and ability to innovate (Davenport & Prusak, 1998). It is based on accumulating experiences and learnings, which build a shared knowledge base that makes it easier to perform tasks and speed up innovation (Nonaka & Takeuchi, 1995; Argote & Ingram, 2000).

Nonaka and Takeuchi point out that knowledge creation starts at the individual level but becomes a collective process through organizational networks. This dynamic is based on human interactions and exchanges, known as knowledge conversion.

Within this framework, four knowledge conversion processes are identified:

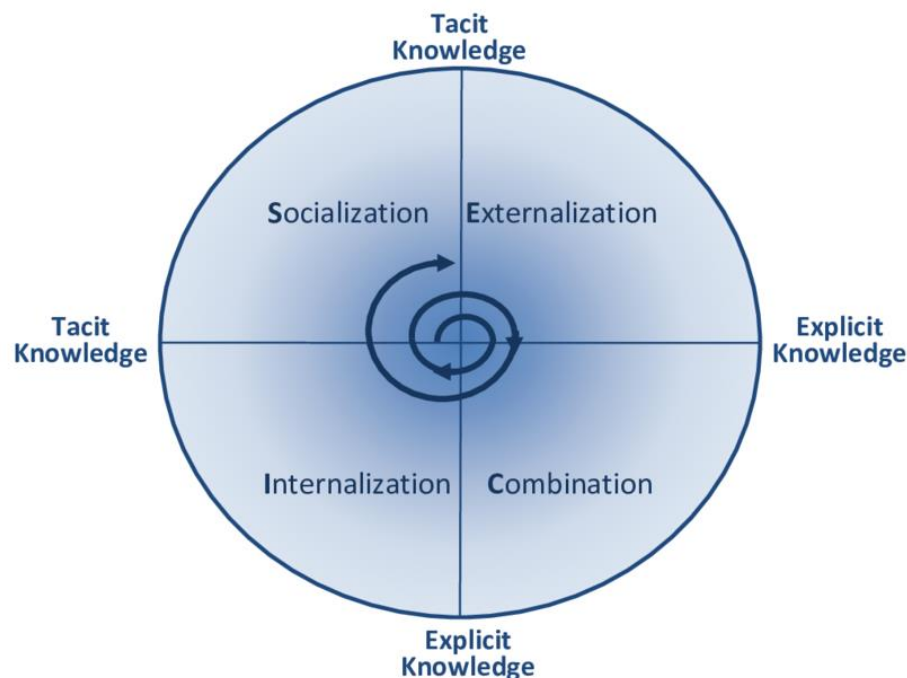


Figure 4 : Nonaka & Takeuchi's SECI Model (1995)

- **Socialization:** Sharing experiences to create tacit knowledge, such as shared mental models and technical skills. Experience is key in this process.
- **Externalization:** Turning tacit knowledge into explicit concepts, often using metaphors, analogies, or models.
- **Combination:** Organizing explicit knowledge by grouping and structuring it using various media (documents, meetings, digital communication, etc.).
- **Internalization:** Converting explicit knowledge into tacit knowledge through documents and learning-by-doing practices.

V. Knowledge Management and Culture

Organizational culture plays a crucial role in the process of creating knowledge, especially by converting tacit knowledge into explicit knowledge (Nonaka & Takeuchi, 1995). Human interaction and social dynamics are essential for encouraging the exchange and creation of new knowledge (Nonaka & Takeuchi, 1995). Companies that succeed in knowledge management tend to cultivate a culture that promotes collaboration, trust, and communication (Nonaka & Takeuchi, 1995).

In addition to Nonaka and Takeuchi (1995), several other authors have also highlighted the importance of organizational culture in knowledge management:

- **Davenport and Prusak (1998):** In their book *Working Knowledge: How Organizations Manage What They Know*, they emphasize that knowledge management is based on the culture of an organization. They believe that an organization should create an environment where knowledge is shared, and where trust, collaboration, and interactions are encouraged. Organizational culture sets the rules and behaviors related to managing and sharing knowledge.

- **Sveiby (1997):** Sveiby also stresses the importance of culture in knowledge management. He suggests that companies that succeed in managing knowledge are those that have a culture that values continuous learning and collaboration among employees. For him, knowledge management is strongly influenced by the values and social structure of the organization.

- **Argote and Ingram (2000):** In their study on the importance of knowledge management in organizations, Argote and Ingram point out that organizational culture plays a key role in absorbing and spreading knowledge. They highlight the importance of relationships and social networks within an organization to facilitate knowledge exchange.

- **Alavi and Leidner (2001):** These authors have shown that organizational culture influences knowledge management. In their review of knowledge management systems, Alavi and Leidner explain that successful organizations foster a culture that supports collaboration and the sharing of information among individuals.

- **Spender (1996):** Spender also highlights the crucial role of organizational culture in knowledge management. He points out that the acquisition, management, and use of knowledge are strongly influenced by the values and cultural practices of an organization's members.

All these authors agree that organizational culture is a key factor for effective knowledge management. A culture that values sharing, collaboration, and innovation can greatly improve knowledge management and speed up the innovation process within organizations.

VI. The Importance of Knowledge Management in the Water Sector

In the Tunisian context, the water sector is particularly vulnerable to challenges such as water shortages, climate change, and natural hazards like floods and droughts. Effective water resource management is therefore a national priority, and it is essential to develop an appropriate organizational culture within public administrations to address these challenges.

Knowledge management, supported by an organizational culture that encourages exchange, collaboration, and innovation, is a strategic tool to better manage water-related challenges (Nonaka & Takeuchi, 1995).

In Tunisia, such a culture would allow for the quick and efficient collection, analysis, and sharing of knowledge about water resource management. It would also make coordination between different parties easier, thus strengthening the response to water crisis situations.

Climate change increases the frequency and intensity of extreme events, directly affecting water management. This calls for a proactive approach to gather, share, and use knowledge effectively in order to make informed, data-based decisions. Deepening our knowledge about the impacts of climate change and integrating digital tools and artificial intelligence is crucial to strengthen the resilience of water management systems and infrastructures (Alavi & Leidner, 2001).

Furthermore, floods and droughts, which are becoming more frequent due to climate variations, require collaborative and inter-institutional management. Inter-ministerial relationships are essential to ensure the rapid sharing of crucial information in times of crisis. Collaboration between ministries, local authorities, and private actors is vital for effective water resource management (Argote & Ingram, 2000).

Thus, by developing a culture of knowledge sharing within Tunisian administrations, the water sector could better face challenges such as water shortages, floods, and droughts while improving water resource management through better coordination, faster decision-making, and continuous innovation.

VII. Knowledge Management Concept:

The concept of Knowledge Management involves understanding the processes and systems that organizations use to create, capture, share, and apply knowledge effectively

The following sections present key KM processes and the role of Knowledge Management Systems (KMS) in supporting these process

VII.1 Knowledge management (KM) processes

Several scholars have proposed conceptual frameworks to structure knowledge management (KM) processes and explain their underlying dynamics. A comprehensive review of these frameworks, each outlining key stages in the knowledge management cycle, is presented in Knowledge Management in Theory and Practice (Dalkir, 2017).

a) Meyer and Zack's Knowledge Value Chain Model (1996)

Meyer and Zack conceptualized the knowledge management process as a knowledge value chain. This model focuses on the production and distribution of information products through a series of sequential activities:

- ✓ Acquisition: Collecting raw data.
- ✓ Refinement: Cleaning, structuring, organizing, and indexing data.
- ✓ Storage/Retrieval: Storing data in knowledge repositories and retrieving it when needed.
- ✓ Distribution: Disseminating knowledge to users across the organization.
- ✓ Presentation/Utilization: Applying knowledge appropriately in operational and strategic contexts.

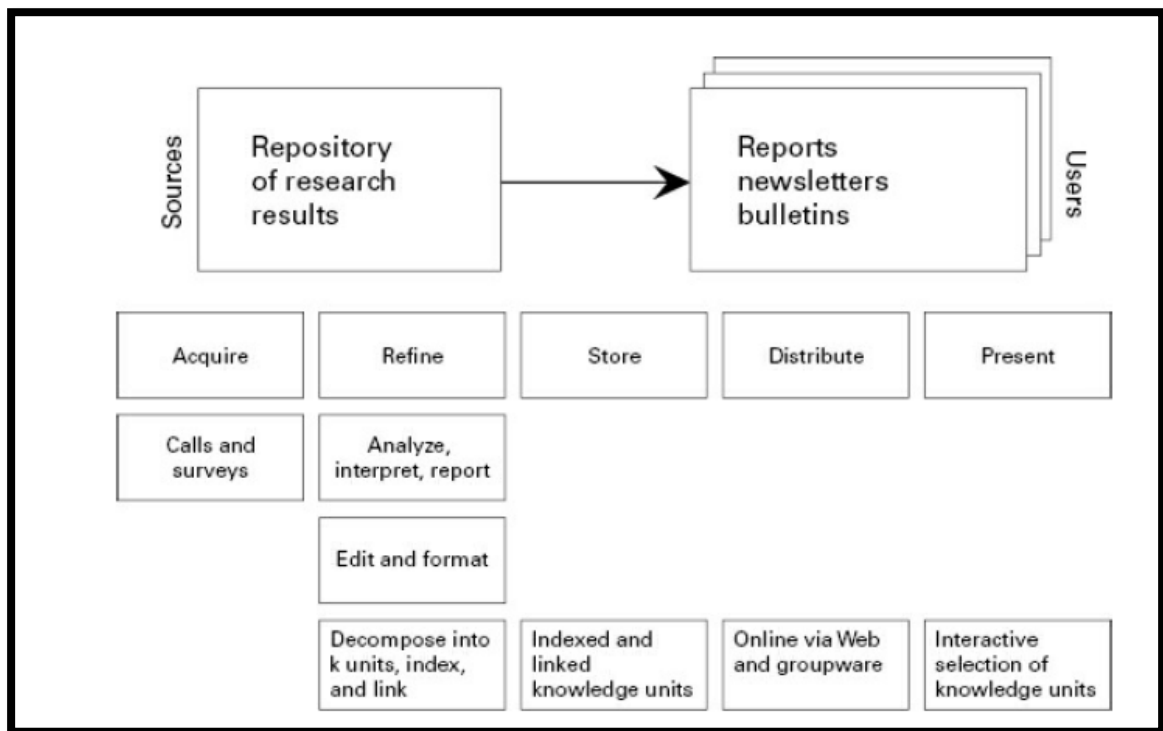


Figure 5 : Meyer and Zack Cycle (Dalkir,2017)

b) Bukowitz and Williams' Dynamic KM Model (2000)

Bukowitz and Williams proposed a dynamic, action-oriented framework centered around decision-making and innovation. Their model incorporates several iterative activities:

- ✓ Get: Seeking necessary knowledge.
- ✓ Use: Applying it to solve problems or foster innovation.
- ✓ Learn: Reflecting on past experiences.
- ✓ Contribute: Sharing knowledge with the collective pool.
- ✓ Assess: Evaluating intellectual capital.
- ✓ Build/Sustain: Maintaining and enhancing knowledge resources.
- ✓ Divest: Eliminating obsolete or redundant knowledge.

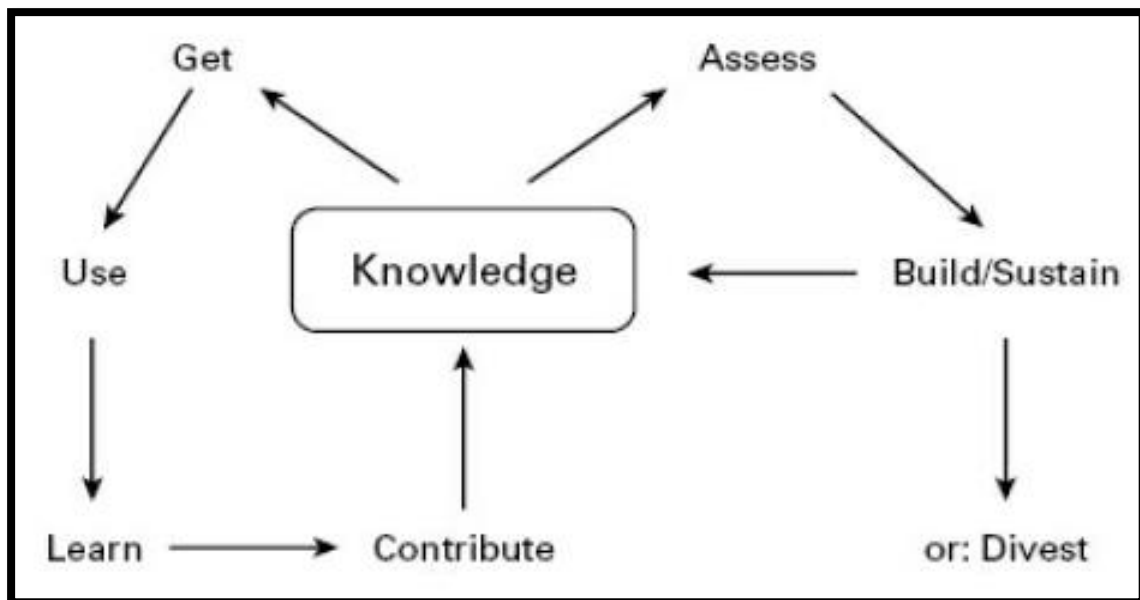


Figure 6 : The Bukowitz and williams KM cycle (Dalkir,2017)

c) McElroy's Knowledge Life Cycle Model (2003)

McElroy introduced a knowledge management framework based on two core processes:

Knowledge Production: Acquiring, generating, and validating new knowledge through individual and collective learning.

Knowledge Integration: Embedding validated knowledge into operational processes and organizational practices.

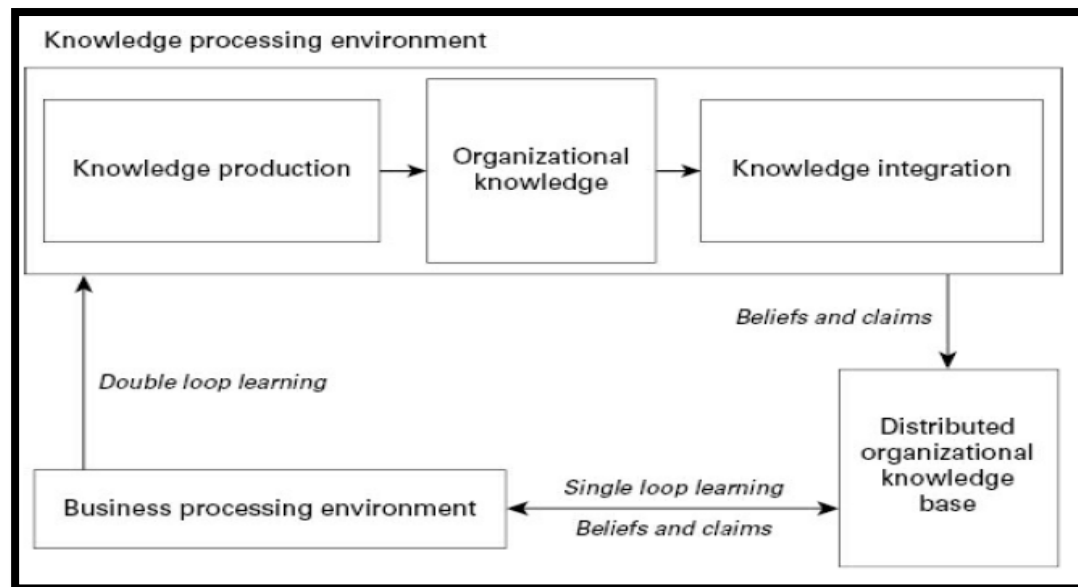


Figure 7 : The McElroy cycle (Dalkir,2017)

d) William R. King's Knowledge Management Cycle (1995)

William R. King proposed a cyclical knowledge management model designed to capture, organize, disseminate, and apply knowledge effectively within organizations. The model involves:

- ✓ Creation and Acquisition of new knowledge internally and externally.
- ✓ Refinement of collected knowledge through organization, structuring, and validation.
- ✓ Integration of refined knowledge into organizational memory systems.
- ✓ Transfer and Sharing through targeted communication and open channels.
- ✓ Utilization within business processes to drive innovation, learning, and operational improvement.

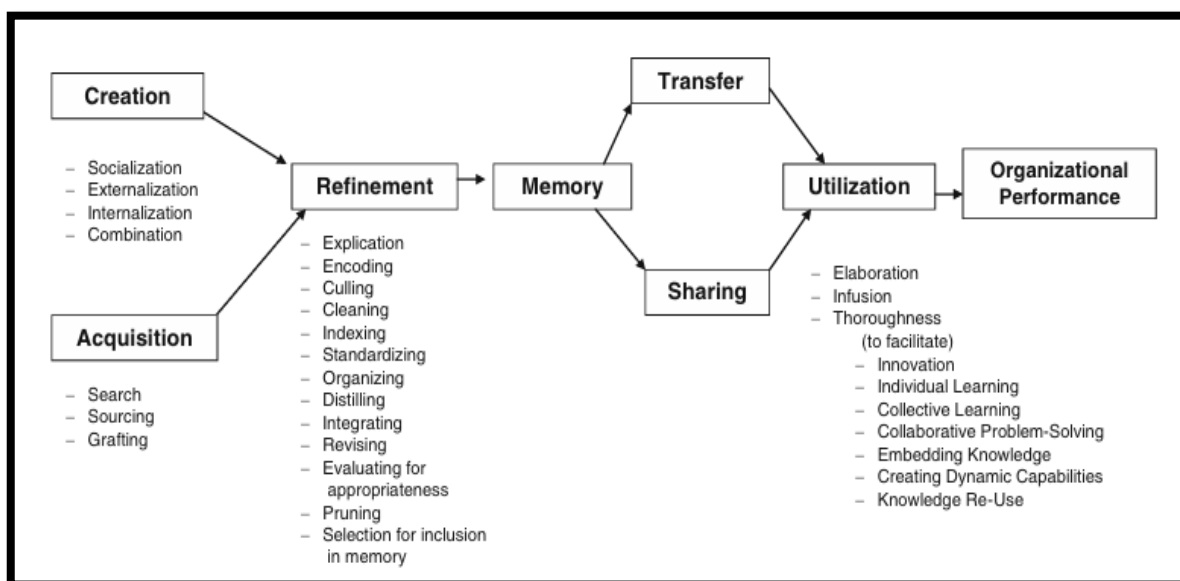


Figure 8 : William R. King's KM cycle (Dalkir,2017)

e) Wiig's Knowledge Management Model (1993)

Wiig emphasizes the construction, retention, and application of knowledge within organizations. His framework comprises:

- ✓ Knowledge Building: Acquiring, analyzing, synthesizing, and organizing knowledge.
- ✓ Knowledge Holding: Storing, integrating, and preserving knowledge assets.
- ✓ Knowledge Pooling: Coordinating and sharing knowledge resources.
- ✓ Knowledge Application: Using knowledge to accomplish tasks and solve problems.

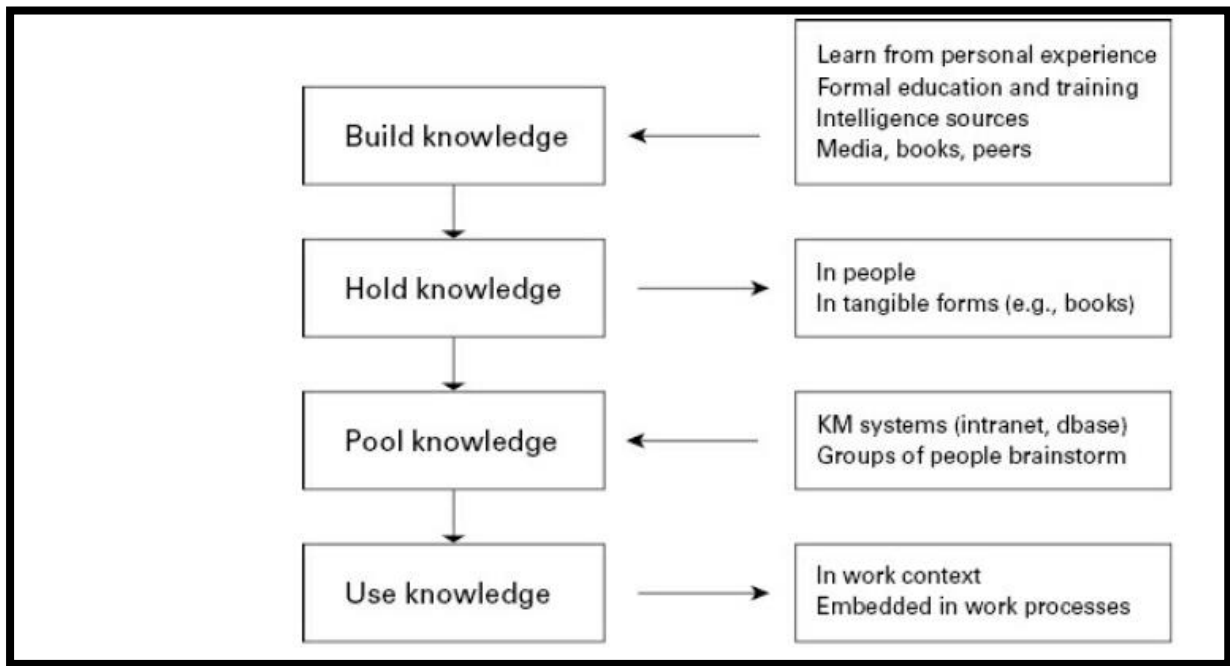


Figure 9 : The Wiig KM Cycle (Dalkir,2017)

f) Carlile and Rebentisch's Knowledge Transformation Model (2003)

Carlile and Rebentisch focus on the management of knowledge across boundaries within organizations. Their model highlights:

- ✓ Acquisition: Gathering new knowledge.
- ✓ Storage: Adding it to existing repositories.
- ✓ Retrieval: Accessing relevant knowledge as needed.
- ✓ Transformation: Adapting and reusing knowledge in new contexts.

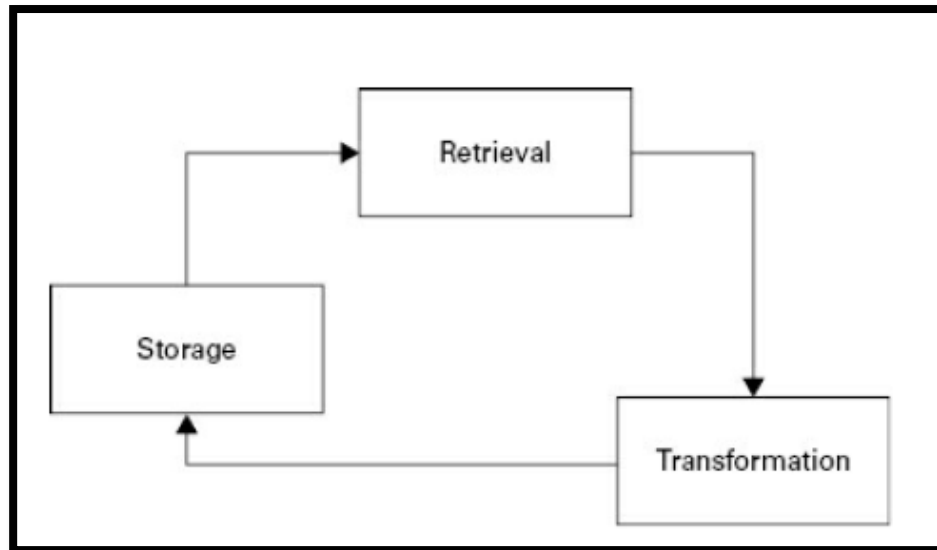


Figure 10 : The Carlilr KM Cycle (Dalkir,2017)

g) Evans, Dalkir, and Bidian's Integrated KM Cycle (2015)

Evans, Dalkir, and Bidian propose a holistic, integrated view of KM processes. Their Knowledge Management Cycle (KMC) includes seven interrelated phases:

- ✓ Identify: Determining whether the required knowledge already exists or must be created.
- ✓ Store: Capturing and organizing knowledge within accessible repositories.
- ✓ Share: Disseminating knowledge internally and externally.
- ✓ Use: Applying knowledge for operational decisions, problem-solving, and innovation.
- ✓ Learn: Reflecting on outcomes and experiences gained through knowledge application.
- ✓ Improve: Refining, updating, and enhancing existing knowledge.
- ✓ Create: Generating new knowledge to address emerging needs.

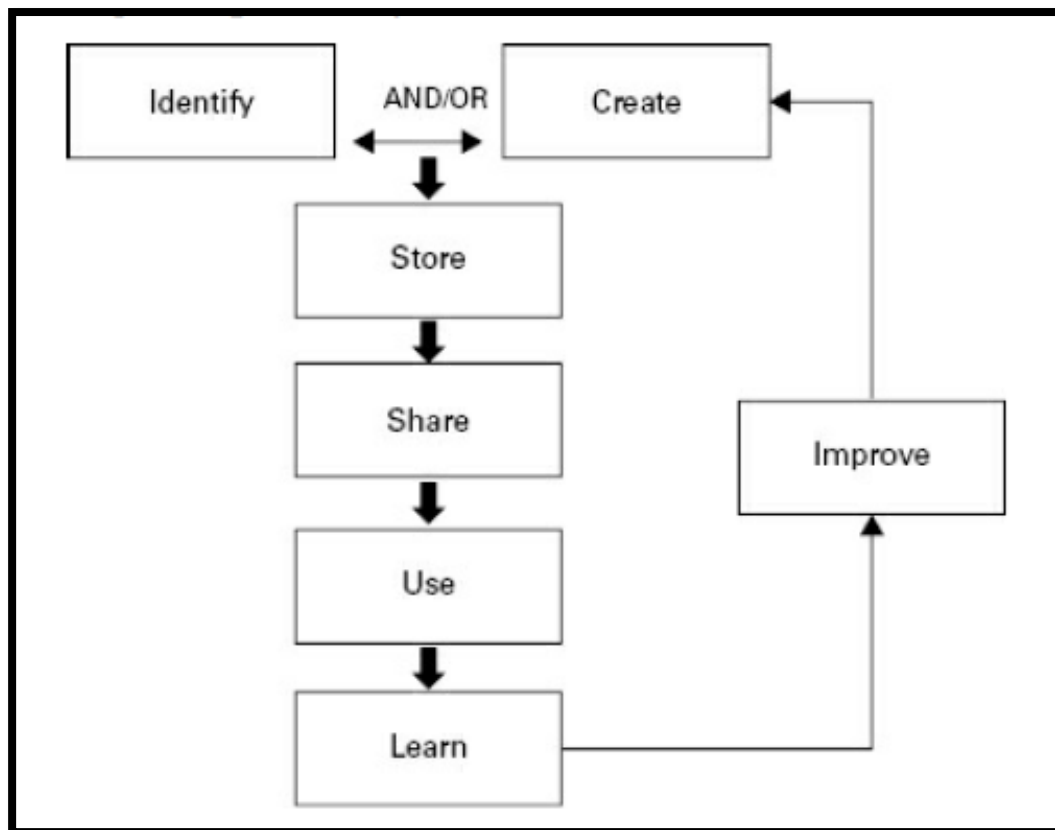


Figure 11 : The Evans, Dalkir and Bidian KM Cycle (Dalkir,2017)

h) Kimiz Dalkir's Knowledge Management Process (2017)

After reviewing the main knowledge management process models proposed by all these earlier scholars Kimiz Dalkir developed an integrated and comprehensive framework in her 2017 book Knowledge Management in Theory and Practice.

Recognizing the recurring stages and complementary aspects of these models, Dalkir synthesized their key elements into a unified KM process model that includes six interdependent activities:

- ✓ Knowledge Identification: Recognizing what knowledge is needed and where it resides.
- ✓ Knowledge Capture: Gathering tacit and explicit knowledge from varied sources.
- ✓ Knowledge Codification: Structuring and organizing knowledge for accessibility and usability.
- ✓ Knowledge Sharing: Disseminating knowledge to the relevant stakeholders.
- ✓ Knowledge Application: Integrating and using knowledge to enhance decision-making, improve processes, and drive innovation.
- ✓ Knowledge Evaluation: Measuring the effectiveness of KM initiatives and their impact on organizational performance

Based on these activities, Dalkir proposes an integrateda KM cycle composed of three main stages:

- ✓ Knowledge capture and/or creation
- ✓ Knowledge sharing and dissemination
- ✓ Knowledge acquisition and application

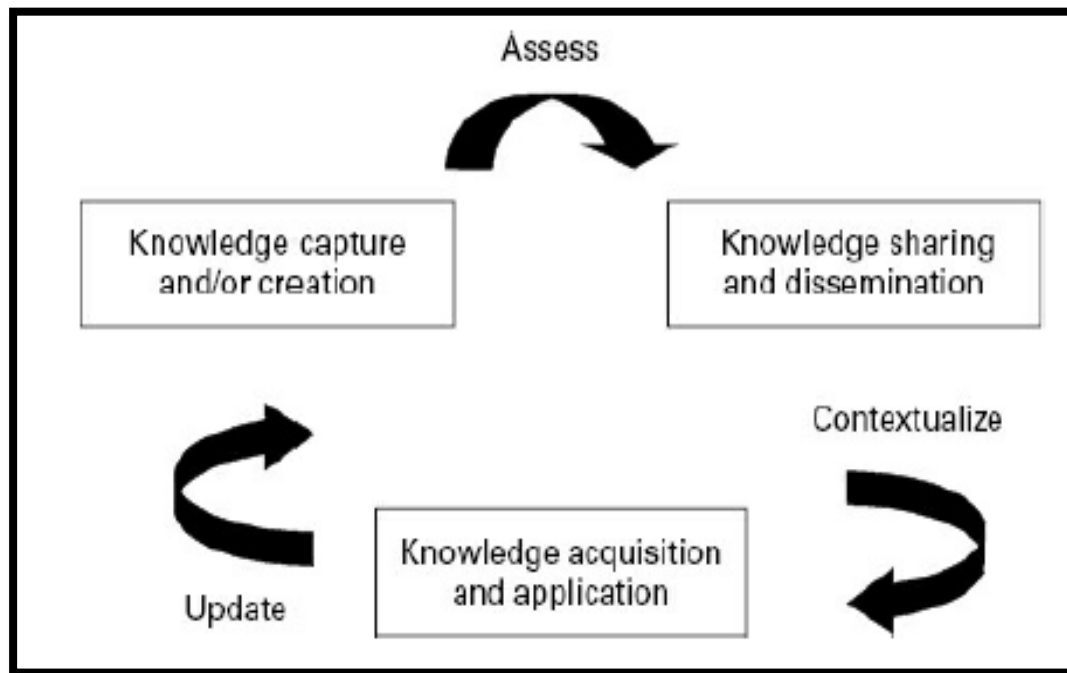


Figure 12 : The Dalkir KM Cycle (Dalkir,2017)

During the transition from capture/creation to sharing/dissemination, knowledge is assessed and contextualized to enhance understanding and application. Contextualization includes linking knowledge to its creators or experts, adapting it to different users, and embedding it into business processes.

the knowledge management cycle, as described by Dalkir, is iterative: Users continuously validate the usefulness of knowledge, detect outdated or irrelevant content, and contribute new knowledge. Through this dynamic process, learning is reinforced at the individual, group, and organizational levels, ensuring the ongoing evolution and improvement of the knowledge base.

VII.2 Knowledge Management System (KMS)

According to Kimiz Dalkir a Knowledge Management System (KMS) is defined as:

"A Knowledge Management System (KMS) is an integrated set of technologies and processes that facilitate the capture, storage, retrieval, sharing, and application of knowledge within an organization. The primary objective of a KMS is to make knowledge accessible to individuals or teams when they need it, in order to make informed decisions, solve problems, innovate, and continuously improve organizational performance."

Dalkir highlights the key components of a KMS as follows:

- ✓ Knowledge repository: A central storage where knowledge assets are categorized and easily accessible.
- ✓ Search and retrieval tools: Effective search functionalities with advanced filters and tagging systems to quickly locate specific knowledge.
- ✓ Collaboration features: Tools such as discussion forums, wikis, and shared workspaces that encourage knowledge sharing and collaborative problem-solving.
- ✓ Access control and security: Mechanisms to protect sensitive information (role-based access, encryption, authentication).
- ✓ Analytics and reporting: Tools to measure the effectiveness of knowledge management efforts, track usage, identify gaps, and make data-driven decisions.
- ✓ Integration capabilities: Connection with other organizational systems such as CRM, ERP, and project management tools.

This definition emphasizes the importance of a structured and organized framework to effectively manage an organization's intellectual assets, making knowledge accessible and usable to enhance overall performance.

In this section I presented the main technologies used in the development of the Rainfall Platform :

The system is designed as a three-tier web application composed of a Frontend Layer , a Backend Layer And a Database Layer.

I had selected each technologie to ensure performance, scalability, and compatibility for managing and visualizing rainfall data at a national level And it is divided into two main categories:

- ✓ Web Application Technologies: used for the frontend and backend development.
- ✓ Database and Data Management Technologies: used for data storage, modeling, and access.

VII.1 Web Application Technologies

these technologies were used to develop the user interface and server-side components of the rainfall platform : and they are divided into two types :

VII.1.1 Frontend technologies:

which manage the visual presentation, user interaction, and data visualization of the platform:

- ✓ HTML5 (HyperText Markup Language version 5) is the standard markup language used to structure web content. It enables the integration of text, images, videos, and interactive elements across multiple platforms. In the rainfall platform , HTML5 structures the visible interface of the user application (World Wide Web Consortium, 2017).
- ✓ CSS3 (Cascading Style Sheets version 3) is a style sheet language used to define the appearance and layout of web pages, including colors, fonts, spacing, and animations. It provides a responsive and visually consistent interface across various devices (World Wide Web Consortium, 2021).
- ✓ Bootstrap 5 is an open-source CSS framework designed to create responsive and consistent web interfaces quickly, using a predefined grid system and reusable components. It facilitates the development of ergonomic and adaptive designs compatible with desktops, tablets, and smartphones (Bootstrap Team, 2023).
- ✓ JavaScript (ES6) is a client-side programming language that allows web pages to become interactive and dynamic. It manages user interactions, client-side logic, and REST API requests to the backend (Mozilla Developer Network, 2024).
- ✓ Chart.js is an open-source JavaScript library that generates interactive charts and graphs from data sources such as JSON or APIs. It is used to visualize rainfall indicators, trends, and historical rainfall data (Chart.js Contributors, 2024).
- ✓ Leaflet.js is an open-source JavaScript library for creating interactive maps using geospatial data.

In this system, it displays rainfall stations and provides a geographic visualization of rainfall indicators (Agafonkin, 2024).

VII.1.2 Backend technologies:

which handle business logic, data processing, authentication, and communication with the database through RESTful APIs.

- ✓ Node.js is a JavaScript runtime environment that enables server-side development using a non-blocking, event-driven model. It executes the backend logic, processes API requests, and facilitates communication between the frontend and the PostgreSQL database (Node.js Foundation, 2024).
- ✓ Express.js is a minimalist framework for Node.js that simplifies the creation of RESTful APIs and web servers. It manages API routes for authentication, rainfall stations, indicators, reports, and user interactions (Express.js Foundation, 2024).
- ✓ JSON Web Tokens (JWT) is an open standard (RFC 7519) used for securely transmitting information between parties as a signed JSON object. It ensures user authentication and access control based on hierarchical roles (technician, governorate, or central administration) (Jones, Bradley, & Sakimura, 2015).
- ✓ PDFKit, ExcelJS, and DOCX : These libraries are used for report generation and data export. PDFKit creates dynamic PDF files, ExcelJS handles Excel spreadsheets, and DOCX generates Word documents from rainfall datasets, ensuring efficient reporting and documentation workflows (OpenJS Foundation, 2024).

VII.2 Database and Data Management Technologies

In this section, the main technologies used for data storage, data processing, and data management within the rainfall platform are presented. These tools form the data layer of the architecture, ensuring that all rainfall information, metadata, and analytical indicators are stored, structured, and accessed in a central database :

- ✓ Python is a high-level programming language widely used for data processing, automation, and backend scripting. In this project, it was used for initializing the database, transforming datasets, and managing data imports and exports (Python Software Foundation, 2024).
- ✓ PostgreSQL PostgreSQL is a powerful open-source relational database management system known for its reliability, extensibility, and compliance with SQL standards. It stores all

rainfall data, station metadata, and computed indicators efficiently (PostgreSQL Global Development Group, 2024).

- ✓ SQL (Structured Query Language) is the standard language for managing and manipulating relational databases. It is used to create tables, insert records, perform analytical queries, and ensure data integrity within the PostgreSQL system (ISO/IEC, 2016).
- ✓ pgAdmin pgAdmin is a graphical administration tool for PostgreSQL that allows developers and administrators to manage databases, run SQL queries, and monitor performance efficiently (pgAdmin Development Team, 2024).
- ✓ PowerAMC (SAP PowerDesigner) known as SAP PowerDesigner, is a data modeling and database design tool used to create Entity-Relationship (ER) diagrams and define database schemas. It helped design the logical and physical models of the PLUVIOM database (SAP, 2023).

All these tools ensure a responsive, secure, and high-performance web application used to managing rainfall data efficiently and interactively.

VIII. Strategic and Innovative Analysis Tools

This section presents the strategic and innovative tools applied in this study to analyze and improve the rainfall data management process. It includes Design Thinking, which provides a human-centered approach to identify user needs and develop solutions, as well as PESTEL and SWOT analyses, which evaluate external and internal factors to guide decision-making and strategic planning

VIII.1 Design Thinking:

Which is a human-centered process that connects user needs, technological possibilities, and economic viability, (**Tim Brown, 2009**).

The Design thinking process has several key characteristics:

- ✓ Iterative: Teams or individuals create, test, and refine ideas to explore many possibilities.
- ✓ Nonlinear: The process can branch in multiple directions or return to earlier stages.
- ✓ Coordinated: A timeframe or set of steps is organized and tracked by a leader, teacher, or coordinator.
- ✓ Collaborative: The team works together to focus on specific areas of exploration.
- ✓ Solution-Oriented: The goal is to develop practical innovations that solve real problems.

The figure below illustrates the three main phases of the Design Thinking process (**Koskinen, 2018**). For him Design Thinking is not only a creative problem-solving process but a structured research method that bridges user understanding and innovation.

He define three phases: Inspiration, Ideation, and Implementation and each phase includes a series of steps that help move from identifying problems to designing and applying effective solutions.

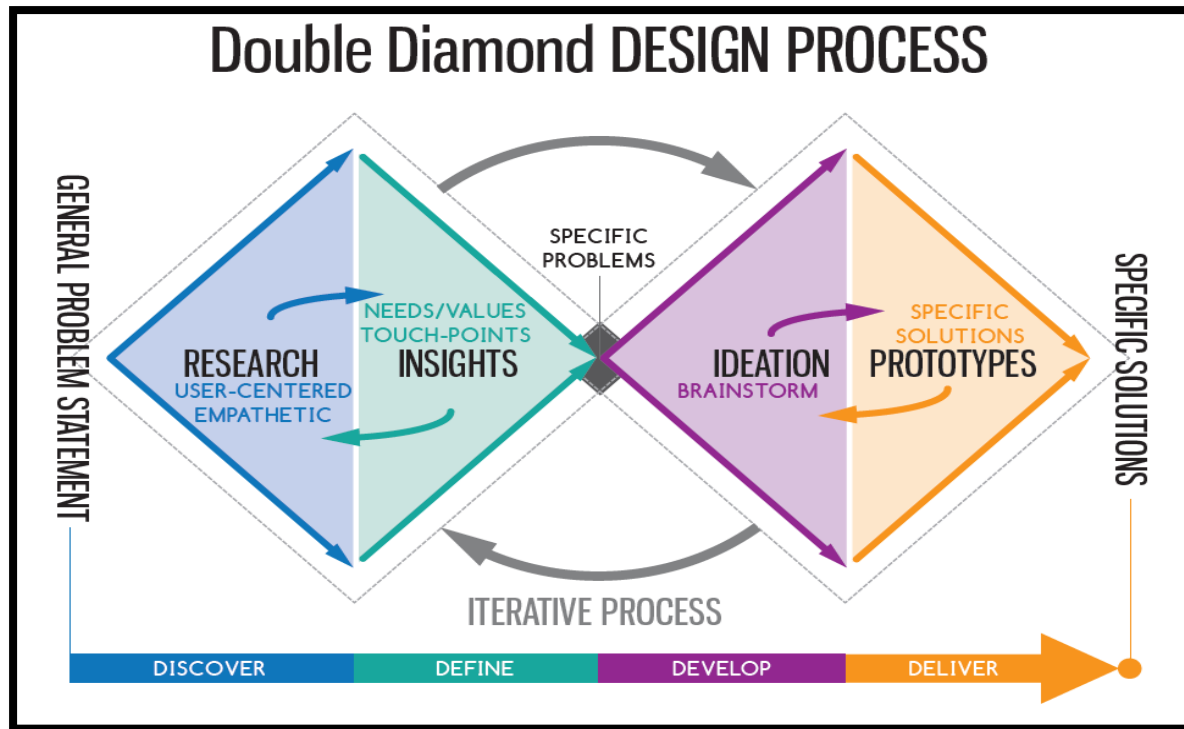


Figure 13 : Phases of the Design Thinking process

1. The Inspiration phase focuses on understanding the problem and need:
 - ✓ It begins with problem articulation and auto-scoping, where the team formulates guiding questions using the *How might we* approach to define the challenge clearly (Brown, 2009).
 - ✓ The next step is Discover and Empathize. This step involves developing a data collection plan to determine what information is needed and how it can be gathered.
 - ✓ Finally, in the Define step, the collected data is analyzed using tools such as personas and journey maps, leading to insight generation and the reformulation of new problem statements (*HMW2*), which guide the next phase.
2. The Ideation phase aims to generate a wide range of creative ideas based on the insights identified earlier. During this stage, the team explores multiple possibilities through brainstorming, collaboration, and co-creation sessions. The goal is to identify innovative, feasible, and user-centered solutions that effectively address the defined challenges.

3. The Implementation phase focuses on transforming selected ideas into practical and testable solutions. This involves prototyping, testing, and refining solutions to ensure they are applicable and aligned with user needs.

Through this iterative process, we can transform ideas to tangible outcomes ready for use.

Lead-user innovation focuses on identifying and learning from lead users who experience needs ahead of the majority of the market. The insights gained from these users are then applied systematically to improve existing solutions, helping companies create innovations that better meet emerging market demands (von Hippel, 1986).

VIII.2 PESTEL Analysis

PESTEL Analysis (Political, Economic, Sociological, Technological, Environmental, Legal) is a framework used to identify and evaluate the major external factors that may influence the project. It provided a better understanding of the broader environment in which the KMS will be implemented (Johnson et al., 2018).

VIII.3 SWOT Analysis

SWOT Analysis (Strengths, Weaknesses, Opportunities, Threats) was applied to assess both internal and external factors specific to the administration. It helped identify the assets of the current data management process, its vulnerabilities,

potential development opportunities, and possible risks to anticipate (Gurel and Tat, 2017).

IX. Conclusion

These four components of Knowledge Management are complementary. A well-defined strategy guides the program, which organizes the processes implemented through a technological system. Their coherent integration enables organizations to transform knowledge into a sustainable competitive advantage.

Chapter III: Analysis & Diagnosis

I. Introduction

Data accessibility has a significant impact on water resource management. To better understand the operational processes, the knowledge flows, and the organization's data management practices, we made an analysis and diagnosis. This work is essential for ensuring effectiveness and serves as a fundamental framework for gaining insights into its performance, strengths, weaknesses, and areas for improvement. This chapter describes the methodology used, the current rainfall data management system, its current situation, and the results.

II. Methodology used:

This section explains the approach and methods used to analyze the current rainfall data management process, identify user needs, and design a practical solution for the DGRE. It details the steps followed, including stakeholder identification, interviews, and the application of Design Thinking principles

The methodology combines the principles of Design Thinking with a user-centered approach to ensure that the new rainfall platform are both creative and relevant, based on a real user needs, and systematically implemented.

In this part I will present all the steps used to analyse the current situation of the rainfall process to collect daily data which are dedicated to the indepth understanding of the data collected process within the DGRE to identify user needs and gather recommendations and to present the final solution proposed by the team to enhance all issues and challenges

The main problem is that the data are scattered and the process is manual which can lead to errors and depends on specific people.

Based on this problem, five several questions are identified to clearly define the challenge:

1. How might we reduce time in our daily process
2. How might we improve data quality and reduce uncertainty
3. How might we valorize our inputs with our benefecils
4. How might we produce decision makers indicators systematically
5. How to improve the process

II.1 Stakeholder Identification

After defining challenges with the director of water surface department, I identified all the stakeholders potentially involved and who can be interviewed for the data collection. They can be divided into the following categories:

- Public users (Citizens, NGOs)
- Researchers (CERTE, ENIT, ...)
- Universities and students
- Ministries (directly influencing institutions such as Agriculture and Transport)
- Other general directorate (DGBGTH , SECADENORD, INM ...)
- Employees of the Ministry,
- administrative staff,
- staff from the Surface Department
- Observers
- CRDA (Regional Commissariats for Agricultural Development)
- General Director
- Director of Surface Water
- Director of water discharge
- technician
- engineer

All stakeholders are presented in the stakeholders map made with the DGRE team

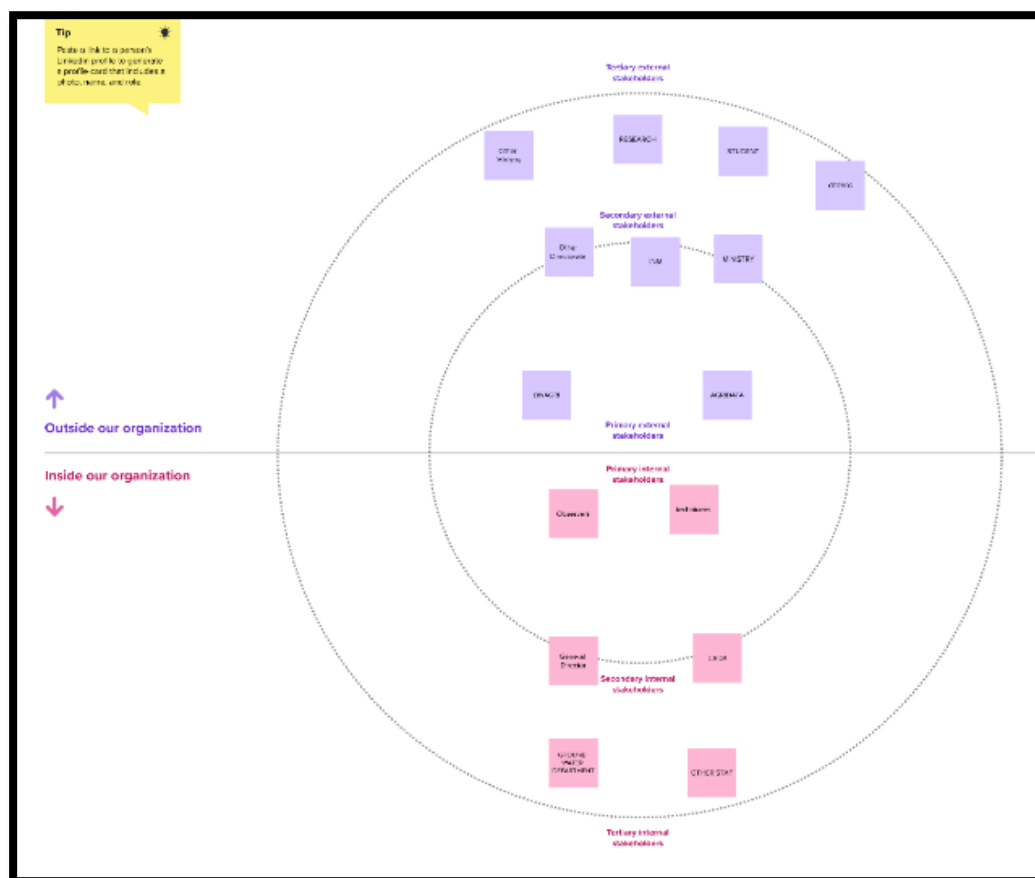


Figure 14 : Stakeholders map

After this I have decided to work in depth with technician , CRDA employee , water surface director and users from the ministry. I generate a list of questions for the CRDA and employee of the ministry to understand the need of the users of the rainfall data . then i made a direct meeting with technician to understand all the workflow and the know_how and finally I have animated a brainstorming session with all the team of the surface water department.

II.2 INTERVIEW

To better understand the needs, challenges, and expectations of actors involved in the management of rainfall data, three types of questionnaires were developed.

-The first questionnaire addressed to technician focuses on the daily processes of data collection and reporting as well as the difficulties encountered and his needs and proposal to improve the process.

Table 1 : Technician's Daily Workflow and Data Processing Assessment

Theme	Question
Data Collection	What are the main steps you follow each morning to collect rainfall data?
	How long does it typically take to collect data by phone from the observers?
	Do you encounter any specific issues when collecting data verbally? If yes, what are they?
	Do some observers occasionally provide estimated values or are not reachable? How do you manage these situations?
Data Entry and Processing	How long does it take to manually enter the data into Excel for all governorates?
	What indicators do you calculate daily? Do you use any personal methods or shortcuts for these calculations?
	Have you ever encountered errors during data entry or calculation? If yes, what were the most frequent causes?
	Which files do you use (names, historical files), and how are they currently organized?
	How do you ensure the traceability of data after entry and processing?
Communication and Publication	How do you transmit the daily rainfall bulletin?
	Do you face any difficulties during the bulletin publication process?
	What kind of feedback do you receive from the end users of the bulletin?
Improvement and Suggestions	Which steps of the current process could be automated or simplified?
	What specific features would you like to have in a future system to make your work easier?
	How can your know how be formally documented to reduce errors and preserve expertise?

The second questionnaire addressed to the director for his positioned at a strategic and managerial level. He define the primary objectives for improving the process, particularly in terms of reducing

costs and time. This part is used to align the system the organization's decision-making needs, by identifying the Key Performance Indicators (KPIs) that the new system must automatically generate.

Table 2 : Director's Strategic Vision and System Requirements Assessment

Theme	Question
Objectives and Challenges	What are the primary objectives you aim to achieve by improving the rainfall data collection process?
	Which steps in the current workflow are the most time-consuming or most prone to errors?
Automation and Time Efficiency	To what extent would you like to automate calculations and the generation of indicators?
	How do you envision the technician's role evolving after the implementation of an automated system?
	What key features do you expect from a new data management system?
Data Management	How would you like historical data to be made accessible and analyzed by your teams?
	What measures should be implemented to ensure data security and prevent loss or corruption?
	How do you foresee the future communication and distribution of rainfall bulletins?
Indicators and Monitoring	What key indicators should the system automatically generate to support decision-making?
	What new skills or training should technicians develop to maximize the benefits of the new system?

The third questionnaire is addressed to the end and their expectations. It guides the development choices for the user interface.

Table 3 : End User Data Usage, Quality, and Access Needs

Theme	Question
Usage and Frequency	How often do you consult the rainfall bulletin?
	For what specific purposes do you use the rainfall data?
Data Quality and Accessibility	Is the current data sufficient for your operational needs?
	Do you face any difficulties accessing or comparing historical data?
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?
	Are you satisfied with the timeliness of the bulletin publication?
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?
	What new features or improvements would you like to see in a future system?

Theme	Question
Suggestions and Improvements	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?
	What improvements could make data consultation and analysis easier for you?

The detailed results obtained from the different questionnaires are presented in Annex 1 for clarity and reference. The main ideas are summarized below with the key findings from the meetings and group work sessions.

III. RESULTS

this section presents the insights from the analysis, including the current workflow of rainfall data collection, the challenges identified, and the proposed improvements. It also summarizes the insights obtained through PESTEL and SWOT analyses to support the development of a centralized and automated knowledge management system

III.1 Daily Rainfall Data Collection Process

To thoroughly understand the current rainfall bulletin and its associated Excel files, multiple meetings were conducted with the technician responsible for preparing the daily reports.

These sessions aimed to capture the complete workflow, including the data collection process, the specific Excel files utilized, and the tacit knowledge required to compute the final values reported in the bulletin:

Each morning at 7 a.m., technician contacts the observers in each governorat by phone to collect rainfall data for the past 24 hours. These values, communicated orally, are then manually entered into an Excel spreadsheet organized by governorate.

Once completed, the table is sent to some users and the National Observatory of Agriculture (ONAGRI) by email for publication on its website in a daily bulletin titled "Nashriya".

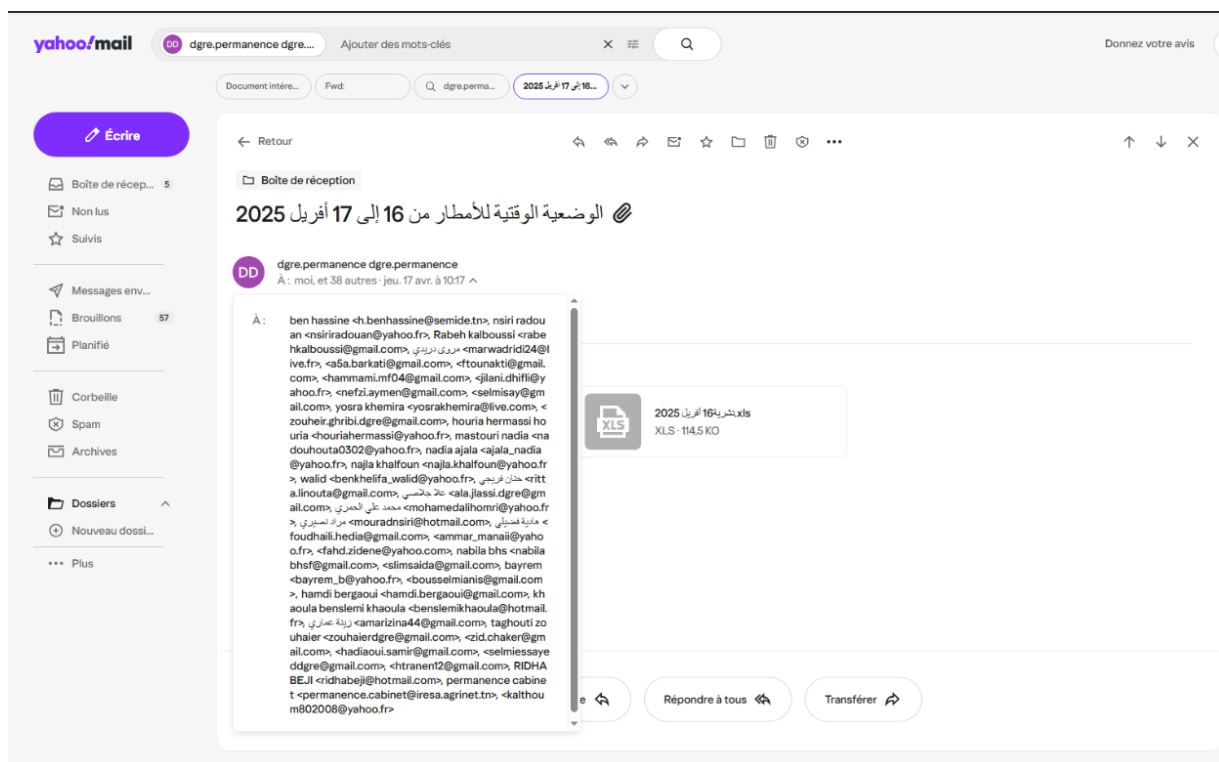


Figure 15 : an Exemple of daily email used to share the bulletin

The bulletin is published in Arabic, as it is the official language of the administration.

A screenshot of the bulletin is shown in Figure 13, providing a visual example of the published data. For reference, the full bulletin for the 23 april, 2025 is presented in Annex 2.

الإدارة العامة للموارد المائية														وزارة الفلاحة والموارد المائية والصيد البحري			
الوضعية الوقتية للأمطار														ملخص توزيع الأمطار حسب المحطات المراقبة :			
2025														23			
أفريل														اليوم			
22														إلى غاية			
الساعة														أفريل			
صباحا																	
النسبة (%)														المعدل			
100*(12/5)=13														100*(9/5)=11			
8-2=10														إيجبي أو سبلي (-)			
إلى اليوم														إلى اليوم			
إلى اليوم														إلى اليوم			
النسبة (%)														النسبة (%)			
100*(6/5)=7														100*(3/2)=4			
6														3			
5														4			
144%														144%			
44.1														44.1			
63.5														63.5			
0.0														0.0			
جندوبة*														جندوبة*			
بوسلم														بوسلم			
غار النماء														غار النماء			
عين دراهم*														عين دراهم*			
طبرقة														طبرقة			
سد بوهرتمة														سد بوهرتمة			
المعدل														المعدل			
85%														85%			
763														763			
124%														124%			
86.9														86.9			
525.5														525.5			
7.5														7.5			
96%														96%			
675.7														675.7			
651.1														651.1			
132%														132%			
71.5														71.5			
94.3														94.3			
0.0														0.0			
باجة														باجة			
مجاز الباب														مجاز الباب			
نفزة *														نفزة *			
سد سيدي سالم														سد سيدي سالم			
أترسي *														أترسي *			

Figure 16 : Extract from the DAILY buletin shared each day

This document is consulted daily by public institutions, farmers, media, and the general public...from the onagri website and It is archived by date, allowing users to access bulletins from previous days as shown in the figure 13.

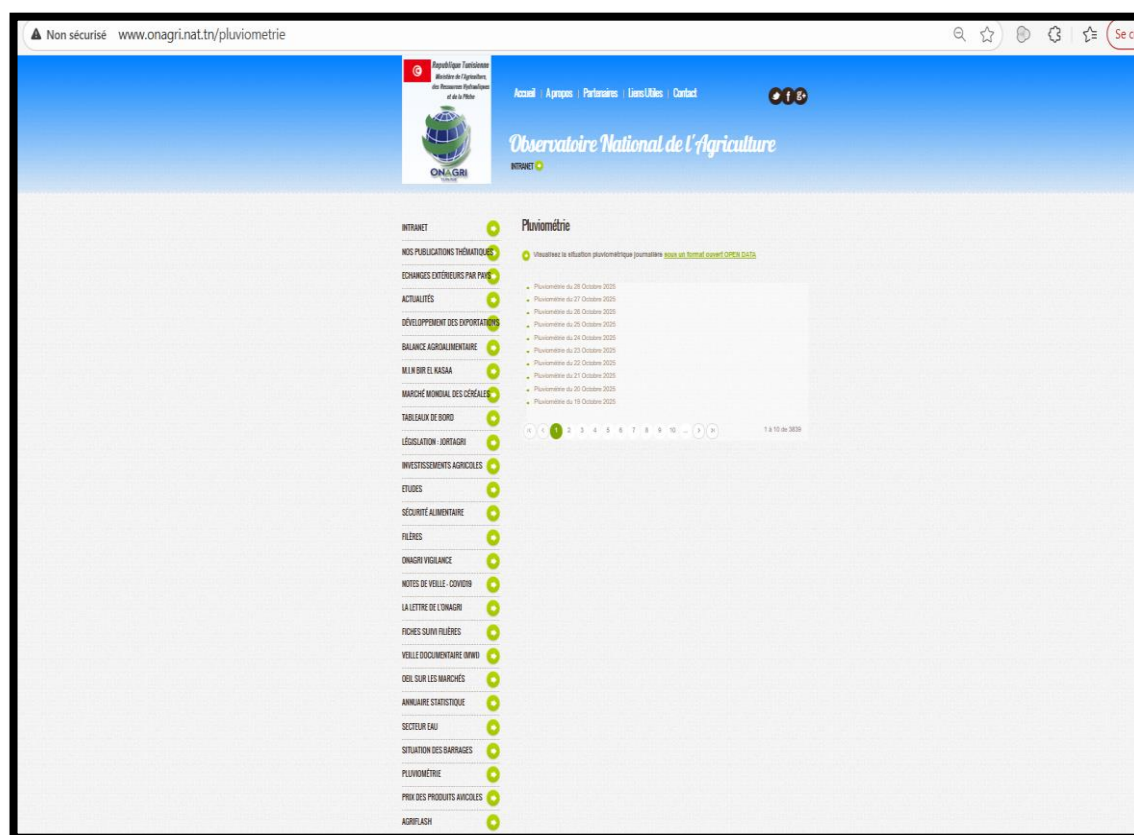


Figure 17 : Rainful excel sheet in the Onagri website

Finally to resume the process made in each day we can find four main steps:

- Manual data collection
- Data transmission
- Manual data entry into an Excel table ,
- Validation
- reporting
- Transmission of the file to ONAGRI,

Figure 14 illustrates the key steps of the current DGRE knowledge management process for rainfall data. The figure outlines each phase, starting with the acquisition of rainfall data, which is collected orally by technicians through phone calls. This information is then manually entered and stored in Excel sheets.

The dissemination phase involves sending the compiled data to ONAGRI and publishing it in the daily bulletin for final utilization by the public, farmers, and decision-makers who consult the data for purposes related to water and agricultural management

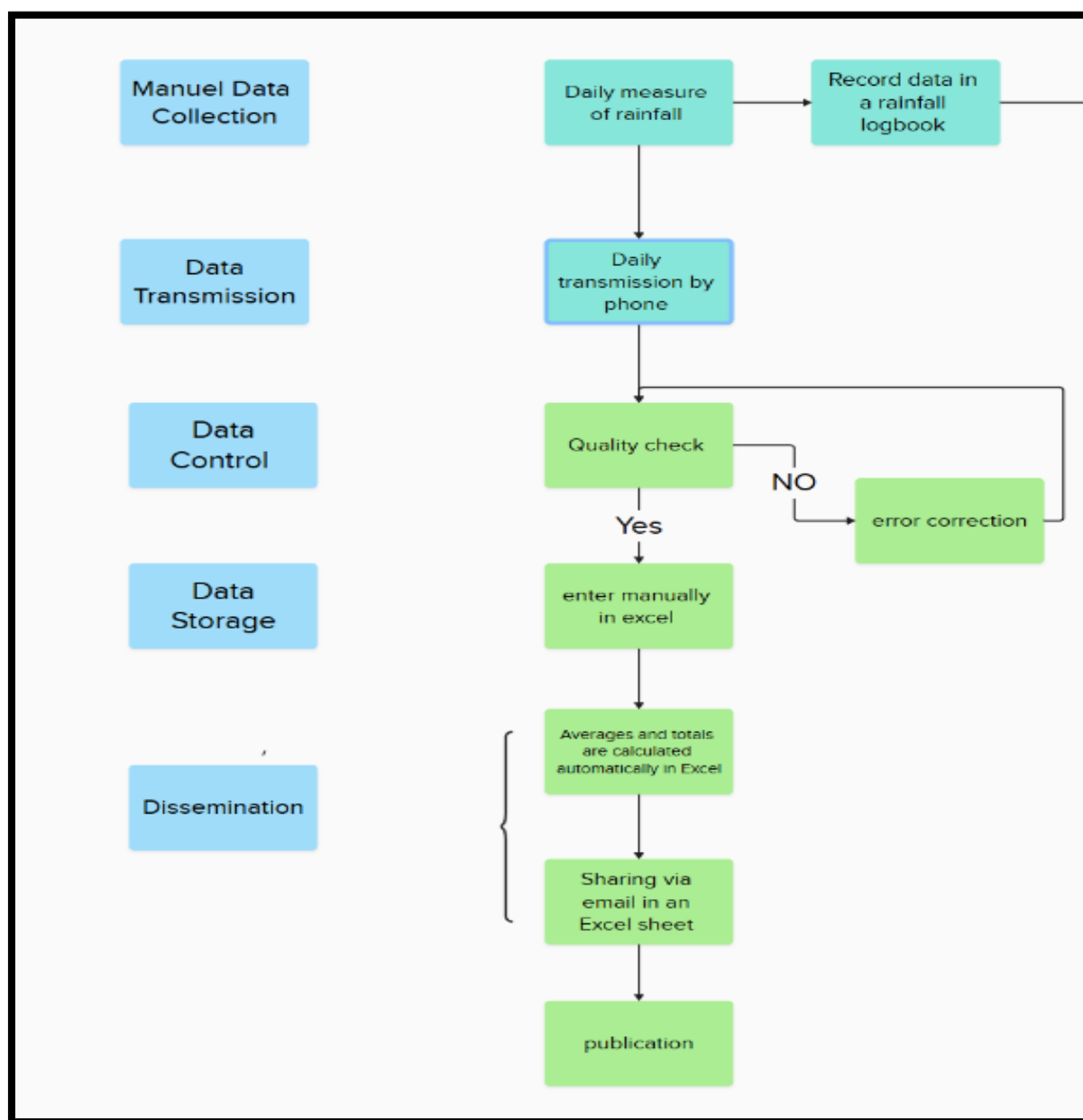


Figure 18 : Daily process of rainfall data

III.2 Understanding the Existing Rainfall Bulletin and Excel Data

It was observed that the data are derived from multiple sources:

- the experience and know-how of field workers and technicians, which is not written down but is essential for the calculation and interpretation of rainfall data.
- Many Excel sheets stored on each technician's computer

-Average Excel shown in the figure 18, which contains daily averages over a historical period of 50 years. The excel shows the average values for each of the 128 stations and covers daily data for the hydrological year from September 1 to August 31, a total of 366 days. The corresponding table consists of 128 rows and 368 columns so in total we have 46 848 values in total.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
9	STATIONS	SEPT																				
10	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
11	JENDOUBA	0.4	1.1	2.1	3.1	3.3	4.0	5.1	5.9	6.7	7.1	7.7	8.4	10.1	12.0	12.2	14.6	15.3	16.3	17.4	19.1	21.1
12	BOU SALEM DRE	0.5	1.3	2.7	3.2	3.5	4.4	5.4	6.6	8.0	8.3	8.6	9.4	10.8	12.2	13.1	14.9	16.1	17.0	17.6	19.3	20.3
13	GARDIMADU DRE	1.6	3.1	3.3	4.5	4.6	4.8	5.3	5.9	7.2	7.3	7.4	8.2	9.0	11.1	11.3	12.2	13.5	14.4	14.8	15.7	16.3
14	AIN DRAHAM SM	2.9	3.5	5.8	7.9	8.5	9.8	12.9	16.3	18.0	18.6	18.9	20.5	22.9	27.3	29.0	31.9	32.7	34.9	38.2	41.2	42.5
15	TABARKA FORET	1.0	1.5	2.5	3.6	4.6	5.6	7.3	9.5	11.8	12.0	12.4	13.7	17.1	19.1	20.2	21.9	24.2	26.8	30.6	33.0	33.5
16	BOU HEURTMA	1.1	1.5	1.9	2.6	3.1	3.6	4.8	5.8	6.2	6.6	6.7	7.2	8.5	11.1	12.3	13.4	15.1	15.8	16.8	17.4	19.0
17	MOYENNE-PER-JOURS	1.3	2.0	3.0	4.1	4.6	5.4	6.8	8.3	9.7	10.0	10.3	11.2	13.1	15.4	16.3	18.2	19.3	20.9	22.5	24.3	25.4
18	BEJA INRAT	0.8	1.2	2.3	3.3	3.9	5.6	7.2	8.5	9.4	10.2	10.4	12.1	13.4	16.3	17.0	18.7	18.9	19.6	20.1	21.6	23.4
19	MEJEZ EL BAB PF	0.4	0.7	1.5	2.1	2.6	3.4	6.8	7.1	8.1	9.7	10.4	11.7	12.9	15.0	16.0	17.4	18.3	18.9	19.9	20.9	22.2
20	NEFZA DELEGATION	0.5	0.7	2.2	3.0	4.1	4.5	5.4	7.5	9.1	9.6	10.0	11.5	13.0	15.1	15.7	18.3	20.1	22.0	23.7	24.6	26.8
21	BARRAGE SIDI SALEM	0.2	0.6	0.9	2.1	2.5	3.5	5.2	6.1	7.0	7.0	8.0	8.8	10.7	14.5	15.5	17.7	18.0	19.3	20.2	21.0	23.3
22	TEBOUSSOUK SM	0.6	1.1	1.8	2.5	2.9	3.8	5.2	6.0	7.7	8.0	8.5	9.3	10.1	12.1	13.3	15.4	16.1	17.2	17.9	19.2	20.6
23	TIBAR SM	1.7	2.3	2.7	3.5	4.1	4.4	5.4	6.0	7.4	7.9	8.2	9.0	9.6	12.3	12.6	15.2	15.9	16.9	17.8	19.2	20.3
24	MOYENNE-PER-JOURS	0.7	1.1	1.5	2.8	3.4	4.2	5.3	6.5	8.1	8.7	9.3	10.4	11.9	14.2	15.0	17.1	17.3	19.9	19.9	21.1	22.8
25	KEF CMA	1.7	3.1	3.3	4.1	4.5	5.1	6.9	8.1	9.2	9.7	10.5	11.3	12.6	13.8	14.3	15.6	16.8	17.5	17.8	18.7	20.0
26	OUED MELLEQUE K13	1.1	1.9	2.5	3.1	3.4	5.5	6.5	7.4	8.5	8.9	9.4	10.3	11.2	13.7	14.3	16.0	16.4	17.4	17.4	18.3	19.1
27	SAKIET SIDI YOUSSEF	1.1	2.2	2.4	3.2	4.5	5.7	7.3	8.9	10.8	11.0	11.6	12.2	13.4	14.6	14.9	16.4	17.4	19.4	20.6	21.7	22.9
28	KALAAT ESSENAM	3.3	3.6	4.3	5.9	6.1	6.4	7.3	8.4	9.8	10.6	11.3	13.1	13.7	15.4	16.3	18.0	18.8	19.8	20.0	21.7	23.2
29	KSOUR ECOLE	1.2	2.2	2.4	3.4	3.5	4.8	5.7	6.6	9.0	9.1	10.2	11.3	13.2	14.9	15.6	17.1	17.9	19.2	21.0	22.7	23.8
30	SERS AGRICOLE	0.6	1.6	1.9	3.4	4.1	4.4	6.9	7.9	9.8	10.5	11.3	11.7	12.9	16.1	16.2	17.8	19.1	19.4	20.9	23.1	25.4
31	TAJROUINE FER D'ETA	1.3	1.9	2.9	4.0	4.2	4.4	6.1	6.9	9.5	11.1	12.0	13.2	14.9	16.3	17.8	18.4	18.9	19.0	20.0	21.8	22.9
32	MOYENNE-PER-JOURS	1.5	2.4	3.2	3.8	4.5	5.2	6.7	7.7	9.9	10.1	10.9	11.5	13.3	15.6	15.6	17.7	17.3	18.9	19.7	21.1	22.4
33	SILIANA AGRICOLE	0.6	1.5	2.0	3.0	3.5	3.8	4.7	5.5	7.6	9.9	10.0	11.8	12.8	15.1	15.9	17.8	20.3	21.7	22.9	23.4	24.9
34	BOU ARADA II	1.0	1.5	2.7	3.6	4.3	4.8	5.4	7.3	8.1	9.2	9.6	10.2	11.1	13.1	14.4	15.0	19.2	20.4	21.7	22.3	24.0
35	KRIB FERME COSSEM	0.7	1.5	2.5	2.8	3.8	4.4	5.9	6.9	8.2	8.8	9.6	10.4	11.5	13.5	15.5	16.4	17.4	18.4	19.4	20.3	22.2
36	ROHIA	1.0	1.7	2.5	3.5	4.3	5.7	7.1	7.6	8.7	9.0	9.7	10.4	11.8	15.3	16.8	18.2	18.6	19.1	20.6	21.1	21.7
37	SAADIA DU BARGOU SM	0.7	2.0	3.4	4.0	5.4	6.4	6.9	8.8	11.5	12.0	12.6	15.4	18.4	19.8	21.8	24.6	26.9	28.2	28.8	29.9	32.4
38	BARRAGE LAKMES	2.1	2.7	3.8	4.3	6.1	6.8	7.8	8.7	10.0	11.1	11.3	12.4	14.0	15.8	18.9	18.7	22.4	24.2	25.6	25.9	27.7
39	MAKTAR PF	1.0	2.7	4.3	5.6	7.0	7.9	8.7	11.1	11.8	13.9	14.3	15.9	18.1	19.3	20.5	21.1	22.1	23.1	24.4	24.7	26.9
40	MOYENNE-PER-JOURS	1.4	2.3	3.0	3.9	4.8	5.7	6.4	8.1	9.4	10.5	11.9	12.3	13.9	16.8	17.4	19.4	21.4	21.9	23.3	23.5	27.1
41	TUNIS CARTHAGE SM	0.9	1.1	1.6	2.9	4.3	5.2	5.9	7.0	7.9	8.7	8.9	9.2	12.3	13.7	15.4	17.9	22.7	24.2	24.5	25.1	27.1

Figure 19 : Extract from data used to calculate indicators

-The previous Excel files to dedicated previous value are calculated by. Each day, the technician sums all daily values from the beginning of the month to obtain the cumulative total.

Table 4 summarizes all the indicators calculated in the Excel sheet used for the daily rainfall bulletin.

Table 4 : Summary of Indicators Calculated in the Daily Rainfall Bulletin

Indicator	Description	Data source	Nbre of column
DAILY rainful	Daily rainfall values	Recorded by field observers	column 2
Monthly cumulative	Sum of daily rainfall from the 1st of the month to the current date	Calculated from the other daily excel sheet	column 3
Monthly average	subtracting the last day of the previous month from the last day of the current month. This value will be updated automatically at the beginning of each month	Average excel sheet	column 4
Percentage of average vs.	The percentage of the monthly average relative to the cumulative total	(Monthly average / Monthly cumulative) * 100	column 5

Indicator	Description	Data source	Nbre of column
monthly cumulative			
period cumulative	The agricultural period cumulative Sum of daily rainfall from th 1 st of September to the current date	Calculated from the other daily excel sheet	column 6
Period averages	The cumulatif average mesured from the 1st of septembre to the some day	The value of the correspondance day in th Average excel sheet	column 7
Percentage of agricultural period average vs. cumulative	The cumulative rainfall of the same month from the previous year	(period average / period cumulative) * 100	column 8
Previous year cumulative month	the cumulative rainfall for the some month of the previous period	the previous year's Excel file for the same date	column 9
Previous year cumulative period	the cumulative rainfall for the some previous period	the previous year's Excel file for the same date	column 10
Difference current vs previous month	The difference between the current and previous monthly	Current monthly cumulative - Previous year monthly cumulative	column 11
Difference current vs previous period	The difference between the current and previous period	Current period cumulative - Previous period cumulative	column 12
Annual average	the annual cumulatif average	value on August 31 from the excel average	column 13
Percentage of annual average vs. cumulative	the percentage relative to the current cumulative	(Annual average / period cumulative) * 100	column 14

This process of the calculation of the indicator is repeated for each station, governorate, and natural region, highlighting the complexity of the daily workflow, the dispersion of data across multiple Excel files, and the risk of data loss or inconsistency.

Following the working meetings and brainstorming sessions held with the DGRE team and based on the responses to the questionnaires, several challenges were identified in the existing process. The main issue is that the current system shows a lack of fluidity and standardization: oral

exchange , double data entry, and lacks synchronization between the collected data and the central DATABASE . These issues result in delays, potential errors, and low data traceability and The calculated data are derived from both raw and historical datasets, as described previously

Based on the in-depth analysis of the current rainfall data management process the HOW MIGHT WE questions are reformulate to guide the ideation and implementation phases:

1. How Might We centralize rainfall data from multiple sources to reduce duplication and improve traceability?
2. How Might We automate repetitive Excel calculations to gain time and reduce human errors?
3. How Might We provide real-time access to rainfall data for all stakeholders?
4. How Might We make historical rainfall data easily retrievable and analyzable
5. How Might We ensure that tacit knowledge from technicians is captured and formalized to automatically updates indicators?

IV. Ideation phase

The insights identified in the previous steps motivated the development of a centralized database to store all rainfall data and a web application to manage data tables, facilitate daily data entry, and automate calculations.

To ensure accuracy, consistency, and efficiency, scripts will be developed to automatically perform all required calculations. These scripts will replace the manual computations currently done in Excel, thereby reducing human error and saving considerable time for technicians.

By implementing these scripts all calculations will be executed automatically as soon as new daily data are entered into the system. This will ensure real-time updates, consistency across all stations, and a clear audit trail for verification. During the Ideation Phase a new automatique process was proposed as shown in the figure 19.

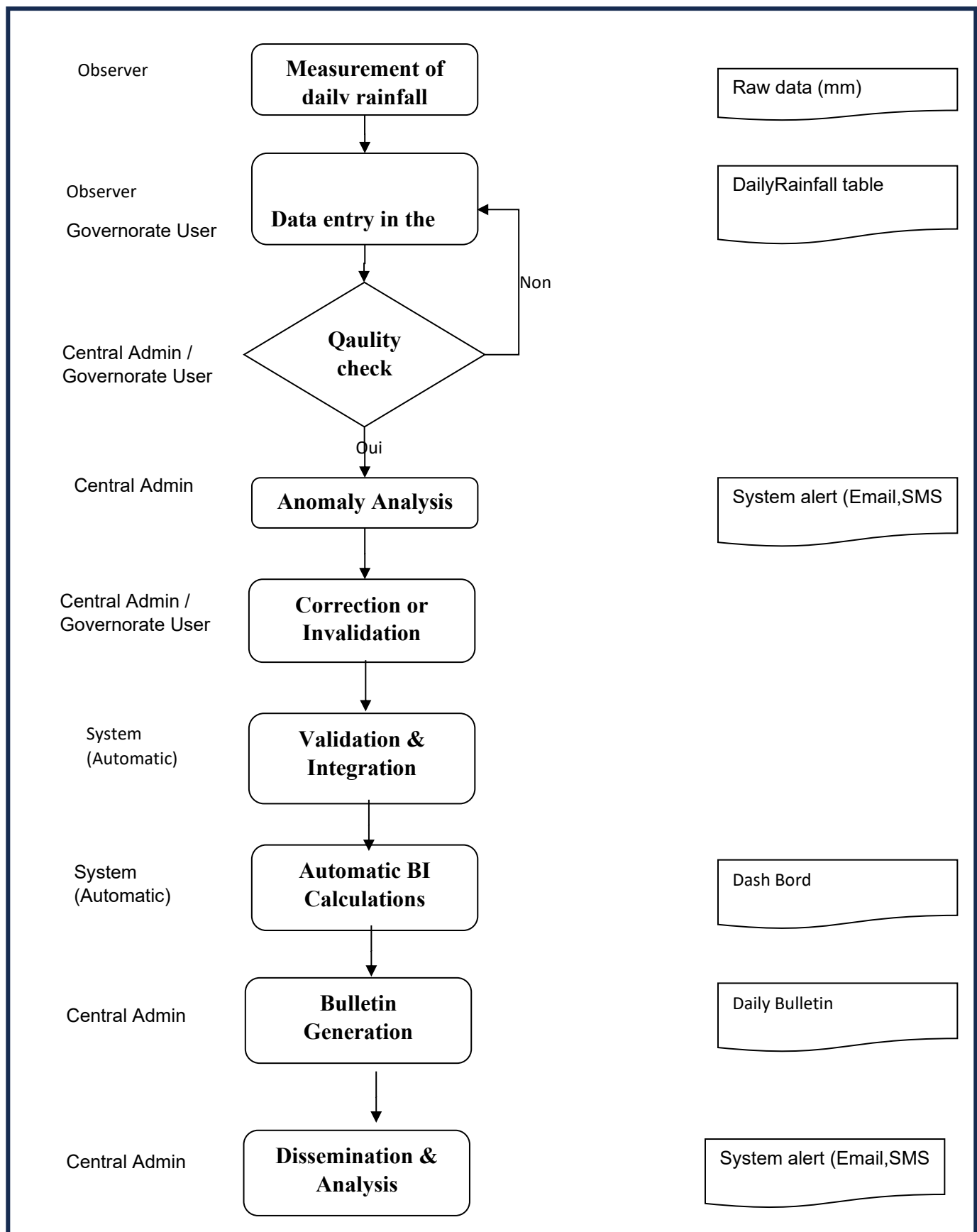


Figure 20 : The new process of the rainful data

1. In the new workflow, observers continue to perform daily rainfall measurements, but data entry is now done directly through a web application. As soon as a record is submitted, it is instantly transmitted to a central database hosted on a PostgreSQL. This completely eliminates the need for manual phone calls.
2. Once the data are received, the system sends an alert confirming that the information has been successfully transmitted to the central level
3. Technician initiating an automatic quality control process. In parallel, automated verification scripts developed analyze the values and detect anomalies, such as rainfall exceeding 1500 mm in a single day or temporal inconsistencies.
4. then an error is detected, the observer receives an alert prompting verification. Once the data are validated by the central user, they are automatically integrated into the calculation process.
5. These automated calculations generate the indicators : daily totals, monthly and period averages, as well as comparisons between the current period and the previous year.
6. Based on these computations, the system produces periodic reports such as the daily bulletin, the monthly situation report, and the annual rainfall summary.
7. The results are then disseminated through interactive dashboards, and the bulletins can be automatically sent by email.

All this information is stored within a national unified database.

The main objective of this platform is to centralize all rainfall data in Tunisia, whether daily, monthly, or recorded by automatic instruments. Another key goal is to automate the calculation indicators, make all calculation rules explicit, and standardize input and output formats to facilitate data exchange between regional offices and the central level.

To analyze this new system, I proceeded by structuring the evaluation around two complementary diagnostic tools: PESTEL and SWOT.

The results of the PESTEL analysis are summarized in Figure 20 , which highlights the key external factors that may influence the development and implementation of the Knowledge Management System (KMS). This summary allows for a clear overview of opportunities and potential constraints in the broader organizational and environmental context:

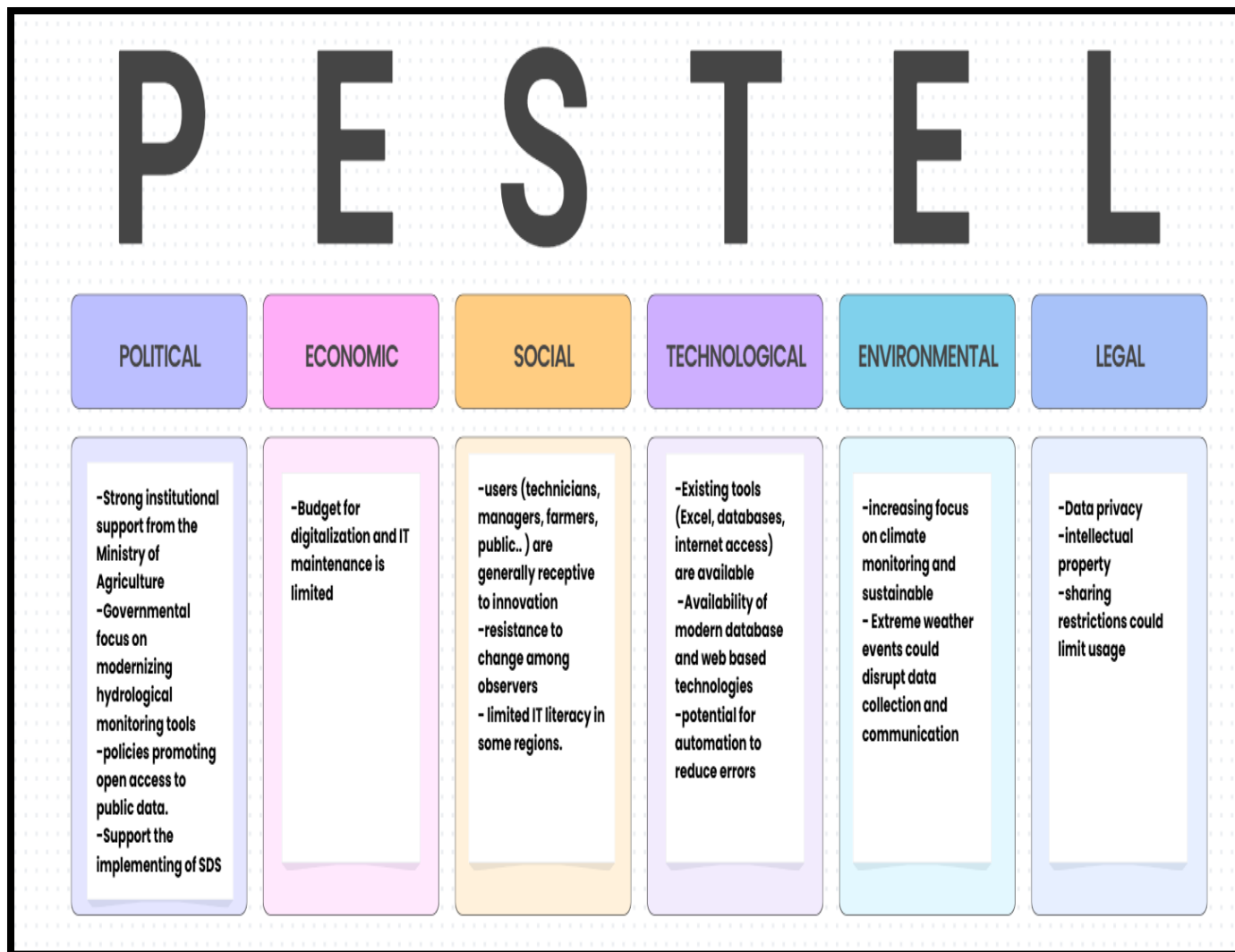


Figure 21 : PESTEL analyses

PESTEL factors indicate a favorable external environment for implementing the new system.

The results of the SWOT analysis are summarized in table 5 providing a clear overview of the internal strengths and weaknesses, as well as the external opportunities and threats affecting the current Rainful data management process. This summary helps identify key assets, vulnerabilities, potential improvements, and risks in implementing the New Knowledge Management System (KMS).

Table 5 : SWOT analyses

Strengths	Weaknesses
-Experienced Technical Staff: Field technicians and operational staff have deep tacit knowledge essential for accurate rainfall data calculation and interpretation -Historical Data Availability -Existing Basic Workflow: The current Excel-based data collection and daily bulletin system provides a foundation for further systematization -Institutional Support: Strong organizational commitment to modernizing hydrological monitoring enhances adoption potential -Clear Knowledge Value Chain -guide process analysis and improvement. -Stakeholder Engagement: Daily consultations and reporting to multiple stakeholders (farmers, public institutions, media) ensure the system meets practical need	- Data collection, entry, calculation, and dissemination are all manual, leading to high labor intensity and potential human errors. - No centralized database; Excel files are scattered across technicians' computers, creating redundancy and inconsistency. - Much of the expertise resides in staff memory, risking knowledge loss if personnel leave or retire. -Data Quality and Traceability Issues: Oral collection and double data entry increase chances of errors, missing data, and discrepancies between published and stored values. -Lack of Standardization: Processes vary across regions, making national comparisons and centralized reporting difficult. -Limited Technical Documentation: Absence of manuals or protocols for data collection and calculations reduces reproducibility and training efficiency. -Limited Analytical Capabilities: Current system lacks advanced statistical analysis, GIS mapping, predictive modeling, and forecasting functionalities. -Geographical Data Gaps: Missing precise station coordinates limit GIS and spatial analysis.

Opportunities	Threats
-Digital Transformation and IT Modernization: National initiatives to improve agricultural monitoring support investment in centralized KMS and web applications. - Integration with Advanced Technologies: Possibility to implement automated rainfall sensors,	Resistance to Change: Field staff and management may resist transitioning from familiar Excel-based workflows to a new system. Budgetary and Resource Constraints: Implementing and maintaining a centralized KMS requires sustained investment in hardware, software, and training.

cloud-based databases, and real-time monitoring systems. -Enhanced Decision-Making: A centralized system can provide accurate, timely information for policymakers, farmers, and researchers. -Alignment with SDGs: Contributes directly to: • SDG 2: Zero Hunger (improved agricultural planning and food security) • SDG 6: Clean Water (better water resource management) • SDG 9: Industry, Innovation, and Infrastructure (digital innovation in agriculture) • SDG 13: Climate Action (climate monitoring and adaptation support) -Improved Data Transparency: Public access to rainfall data can enhance trust and collaboration with stakeholders.	Technical Risks: System failures, cybersecurity vulnerabilities, or integration issues with legacy infrastructure could disrupt operations. - Data sharing policies, privacy rules, or bureaucratic hurdles could limit full KMS implementation. -Dependency on Human Oversight: While automation reduces errors, mismanagement or insufficient training could still result in operational failures. -Rapid Technological Change: Emerging technologies may require continuous upgrades, leading to potential obsolescence of the system.
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V. Conclusion and Recommendations

The implementation of a centralized database and a web-based knowledge management system is essential to address these challenges. The proposed system represent a major advancement in the digitalization of rainfall management in Tunisia.

This initiative aligns with the modernization of hydrological monitoring tools, improving reliability, accessibility, and operational efficiency.

In summary, addressing the identified weaknesses through a robust system will not only enhance the quality and traceability of rainfall data but also strengthen the capacity for informed water management at both regional and national levels.

Chapter IV: the Implementation fo the new RAINFUL managemet system

I. Introduction:

This chapter focus on the implementation phase of the new rainfall system. It includes the final collection of data, the development of the centralized database, and the creation of the fully operational web platform.

The implementation phase is dedicated to putting the system into practice, it aims to integrate all steps of the rainfall management workflow in a single and automated process.

In the following sections, I will present the platform architecture, the database design, and the functional modules that were developed.

II. The Data Base

This section presents the methodology used to design the centralized database for the rainfall management system:

II.1 The Methods used

Designing an information system requires methods to create a model on which the system will be based. Modeling involves creating a virtual representation of reality that highlights the points of interest.

This type of method is called analysis. Several analysis methods exist, with the most commonly used being the **MERISE** Method.

II.2 Presentation of the MERISE Method

According to Jean-Luc Baptiste, (2024) MERISE is a methodology for designing, developing, and implementing information system projects. It is used to facilitate the design of a functional information system and it is based on the separation of data and processes, which are represented through multiple conceptual and physical models

The design of the information system proceeds in phases, each building upon the results of the previous one, ultimately resulting in a functional system that reflects physical reality. Since data and processes are separated, it is important to ensure their consistency, verifying that all necessary data for the processes are present and that no unnecessary data exists

This succession of steps is called the abstraction cycle for information system design:

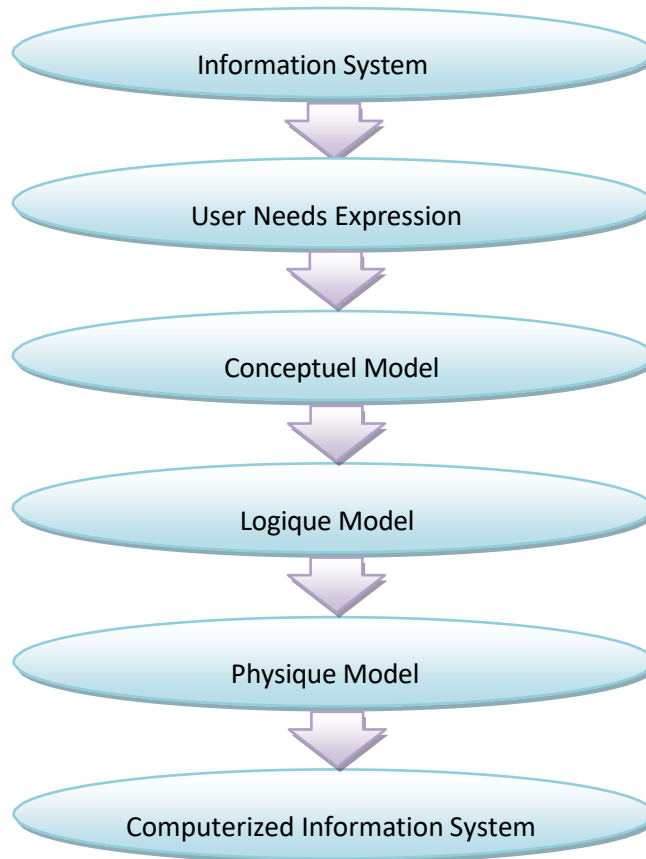


Figure 22 : Abstraction cycle of the information system (Baptiste,2024)

The requirements specification stage focuses on defining the expectations for the automated information system. This includes :

- Taking inventory of the elements required by the information system, and
- Defining the system boundaries in consultation with future users.

This initial phase leads to the creation of the Conceptual Communication Model (MCC), which defines the information flows to be considered.

In the second step the Conceptual Data Model (MCD) is developed, It is used to describes the structure of the data, the relationships between entities, and the business rules and constraints that must be respected.

Based on the MCD, the Logical Data Model (MLD) is constructed to translate the conceptual design into a logical representation that can be implemented in the database

Finally, the Physical Data Model (MPD) specifies the concrete implementation details, such as table structures, indexes, and storage organization, taking into account the chosen DBMS and technical constraints.

At the organizational level, the Organizational Process Model (MOT) defines the processes and constraints imposed by the operational environment including organizational, spatial, and temporal aspects.

The logical level expresses the choices regarding IT resources and means, while the physical level translates these decisions into technical implementations that reflect the specificities of the system architecture.

III.2.1 Conceptual Data Model (MCD)

To automate the processing of information corresponding to the real world, it is necessary to define appropriate representations of different objects and the events connecting them. These relationships are established based on the flow of information circulating between different entities.

The Conceptual Data Model (MCD) represents all the data stored in an information system without considering technical, organizational, or storage/usage optimization aspects. Its goal is to provide a static description of system objects and their associations, which serves to define the files or database structures required for the application (Baptiste, 2024).

Concepts of the Conceptual Data Model

Key concepts in the MCD include:

- ✓ **Property:** The most basic information used to derive entities and their associations.
- ✓ **Entity or Object:** A class of concrete or abstract objects that is relevant for management or business processing. Entities are identified in vocabulary by their “names.”
- ✓ **Association:** Defined by concatenating the identifiers of the entities it links.
- ✓ **Cardinality:** Represents the minimum and maximum number of participations of an entity in an association, expressed as a pair (minimum cardinality, maximum cardinality).

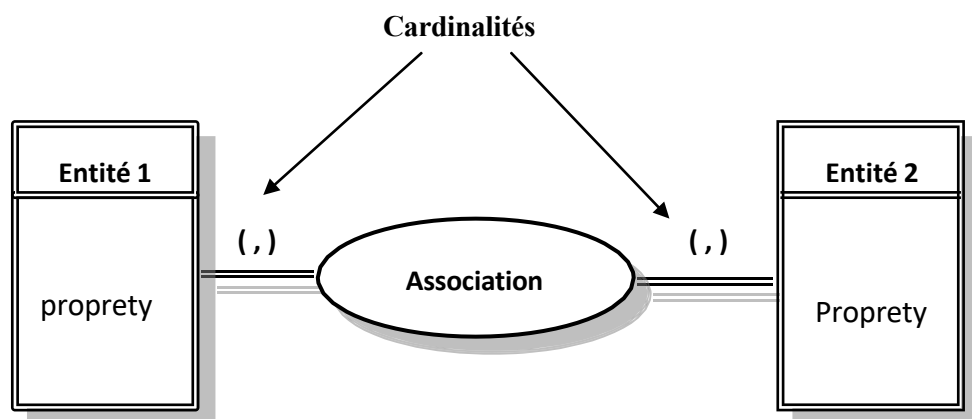


Figure 23 : Graphical representation of the MCD

Table 6 : Cardinality Table

Participation	Optional	MANDATORY
Unique	0,1	1,1
Multiple	0, n	1, n

III.2.2 Logical Data Model (MLD)

The Logical Data Model (LDM or MLD) is the next step in database design, translating the conceptual model into a structure suitable for a database system. It specifies tables, columns, keys, and relationships. For instance, in the rainfall system, the “Station” entity becomes a table with columns for station ID, name, and location. Relationships between entities, such as which observations belong to which station, are implemented using primary keys and foreign keys. The logical model also defines rules to maintain data consistency and integrity. Unlike the physical model, the LDM remains independent of the specific database technology. This stage ensures that the data can be efficiently managed and queried once implemented (Baptiste, 2024).

- ✓ Tables representing entities.
- ✓ Columns representing attributes.
- ✓ Primary keys and foreign keys for relationships.
- ✓ Rules for transforming relationships (e.g., n-m becomes an association table)

III.2.3 Physical Data Model (MPD)

The Physical Data Model (PDM or MPD) describes how the database will be technically implemented in a chosen database management system. It includes the specific table structures, data types, storage organization, indexes, and constraints needed to optimize performance and maintain data integrity. For the rainfall platform, the MPD would define how daily rainfall values are stored, how automated calculations are performed, and how historical data is archived. This model translates the logical design into a practical and optimized system that can be implemented and maintained. The physical model is critical for ensuring that the system performs efficiently and reliably under operational conditions (Elgourari, 2024).

the chosen DBMS. It includes:

- Table structures, data types, and sizes.
- Indexes and storage optimization.
- Constraints, triggers, and rules for performance

Together, these three models form a step-by-step approach to database design. The conceptual model identifies the what, the logical model defines the how in theory, and the physical model defines the how in practice.

This progression guarantees that the system is well-structured, meets user requirements, and is technically feasible. By following this methodology, the rainfall management platform can achieve accurate, efficient, and reliable data processing across all regions of Tunisia (Baptiste, 2024; Elgourari, 2024).

II.3 Choice of Database

To implement this design, **PostgreSQL** was chosen for its high performance in managing both structured and geospatial data. PostgreSQL stores hydrological data, station information, and user management data, and serves as the primary data source for analytical indicators. Its support for geospatial data allows the integration and visualization of information on maps through extensions such as PostGIS (Obe & Hsu, 2015).

PostgreSQL is an advanced, open-source relational database management system (RDBMS) known for its robustness, scalability, and compliance with SQL standards (Stonebraker et al., 2018). It supports complex queries, spatial data, and ensures transactional integrity through ACID compliance.

The database is managed using **pgAdmin**, an open-source graphical administration tool for PostgreSQL. pgAdmin provides a user-friendly interface for database design, query execution, monitoring, and management of both standard and spatial data (pgAdmin Development Team, 2023).

II.4 Database Design

By applying the MERISE methodology, the desired application database was designed. The physical data model, created using **PowerAMC**, is shown in Figure 22.

To implement this database, PowerAMC was used to design the model following MERISE principles, and PostgreSQL was selected as the database management system. The model consists of several groups of tables covering the entire data cycle, from the geographical localization of stations to the generation of analytical reports.

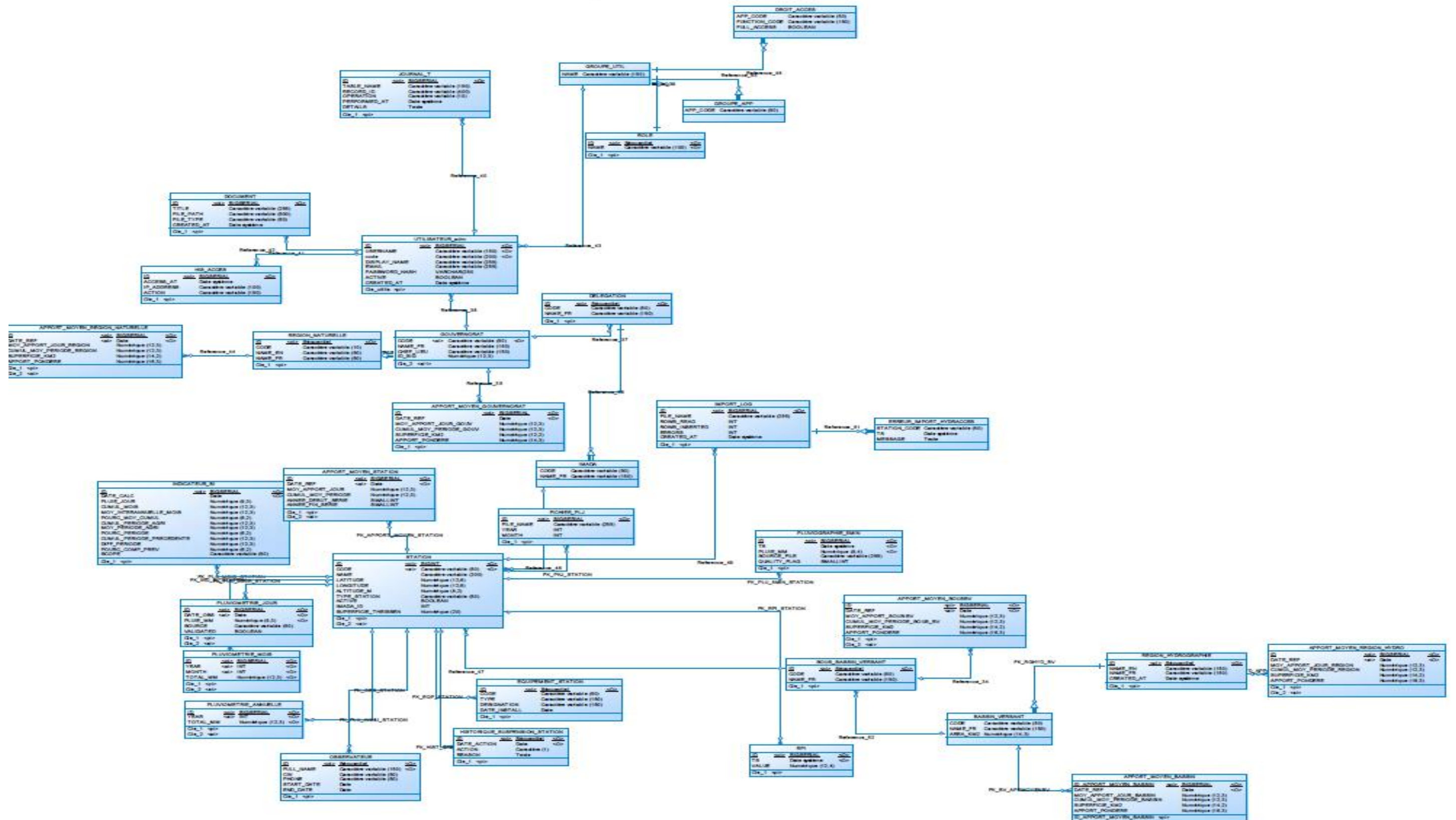


Figure 24 : Conceptuel Model of the PLUVIOM database

II.5 Database Developpement

to developp the data base I used the sql query tool of pg admin to create a complete PostgreSQL database designed for a national platform that integrates rainfall data management across different scales, allowing future expansion to cover all levels of data management.

Figure 23 shows an exemple of the script used to develop the gouvernorat table and the average table. All other tables were created as the same method.

Annex 3 contains all the tables along with their characteristics(columns, keys, and parentchild relationships)

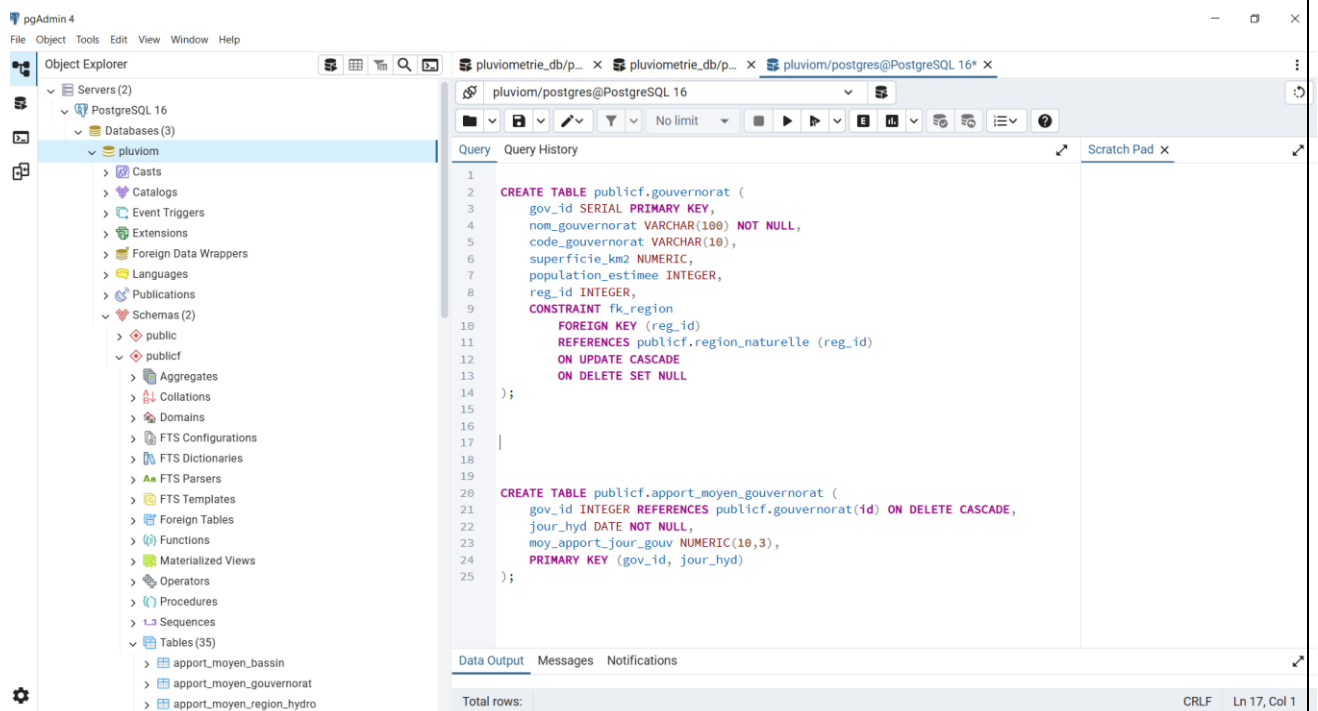


Figure 25 : Exemple of script used to develop table

After executing all scripts required for creating the database tables, the next step consisted of defining the referential integrity constraints to implements a the parent child relationship between table. It enforces referential integrity and guarantees that the data structure remains consistent across both

At this stage, primary keys, foreign keys, and validation rules were applied to preserve the correctness of relationships between entities. The constraint shown in the figure 25 below establishes a referential link between the STATION table and the corresponding APPORT_MOYEN_STATION table to ensure that every recorded average contribution is always associated with a valid station.

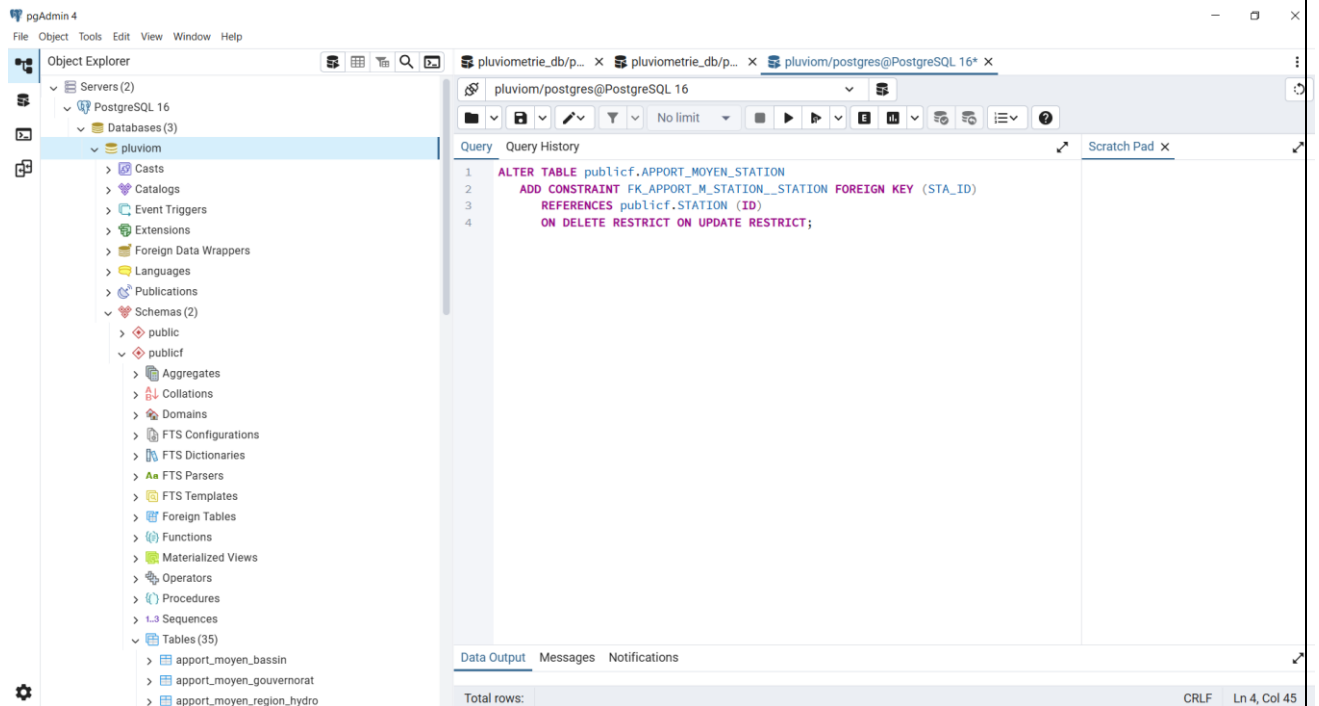


Figure 26 : Exemple of script used to develop constraint

After running the necessary scripts I obtained the main entities, their attributes, and the primary keys, and foreign keys as shown in the figure 24

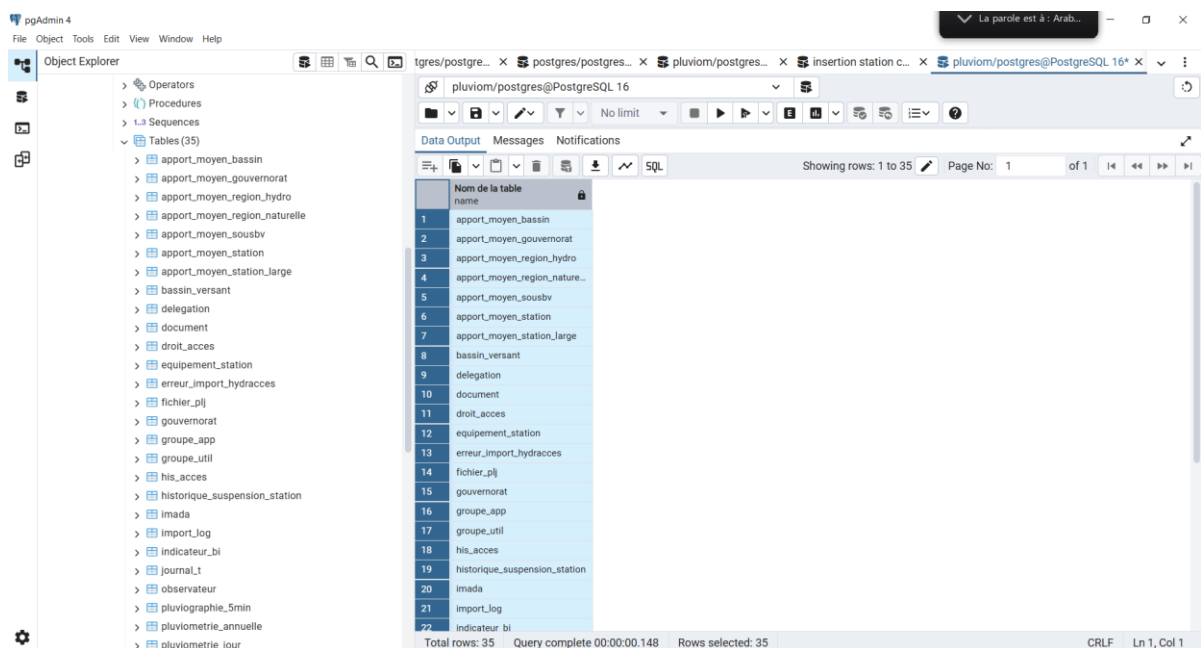


Figure 27 : Tables included in the Data base

The database I developed contain the entire value chain of rainfall data and consists of 35 tables. It includes daily data management and the generation of daily bulletins, which is the focus of this work, as well as annual data management and the production of yearreport as shown in the figure 27.

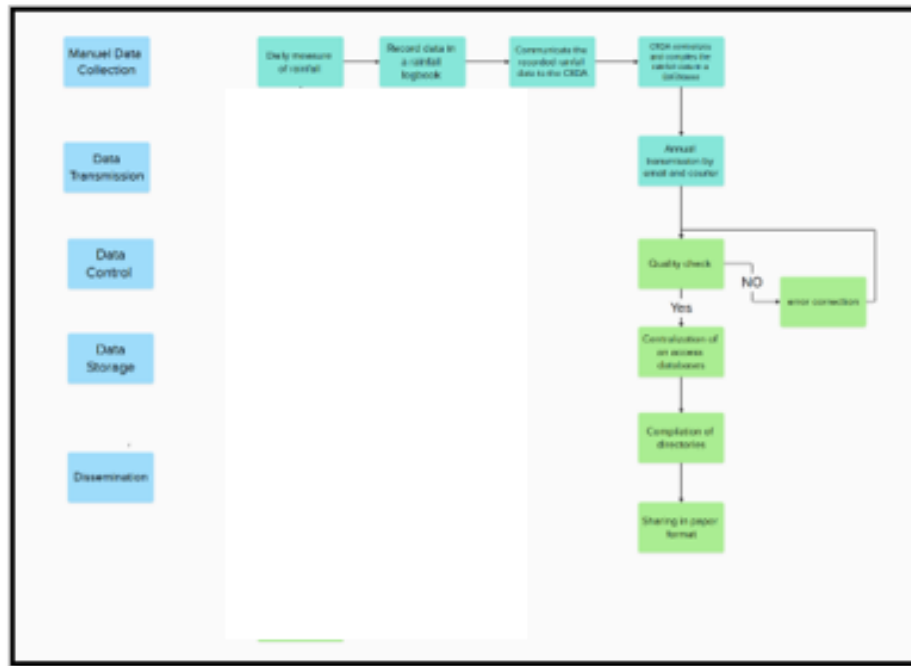


Figure 28 : Annual data management

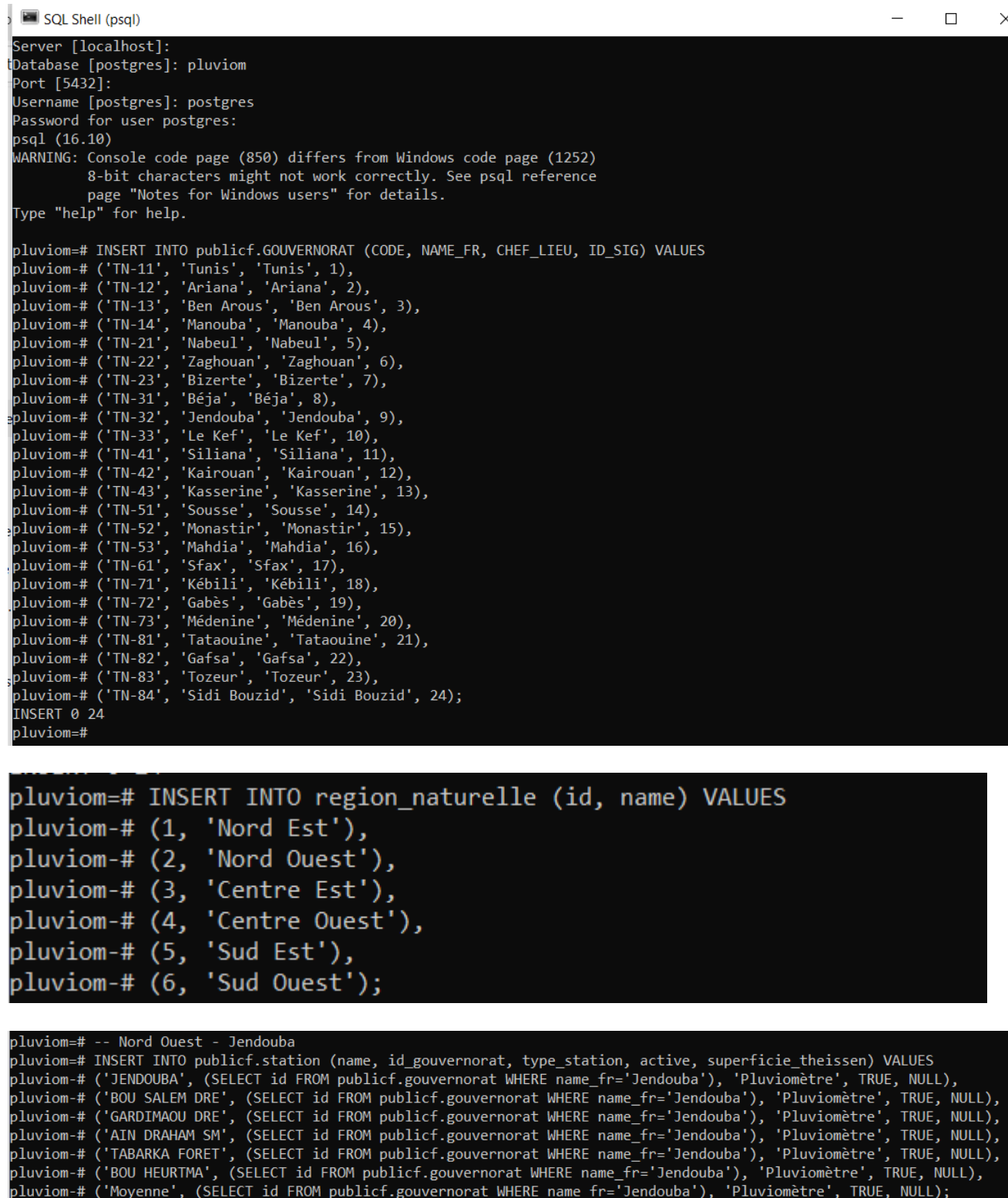
The goal is in my perspective of this work to upgrade the application developed in this master's project into a comprehensive national platform for rainfall data management, integrating all levels of data management in future work.. For the development of the platform, I created APIs that use the following tables:

Table 7 : Tables connected to the system

Number	Table noun
1	station
2	gouvernorat
3	region_naturelle
4	pluviometrie_jour
5	apport_moyen_station
6	apport_moyen_gouvernorat
7	apport_moyen_region_naturelle
8	utilisateur_adm

II.6 Data Integration and Migration

After creating the database, the next step was to populate it with the existing data. To fill the tables for stations, governorates, and natural regions, we used SQL scripts to insert the data directly, as illustrated in Figure 27.



```
SQL Shell (psql)
Server [localhost]:
Database [postgres]: pluviom
Port [5432]:
Username [postgres]: postgres
Password for user postgres:
psql (16.10)
WARNING: Console code page (850) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

pluviom=# INSERT INTO publicf.GOUVERNORAT (CODE, NAME_FR, CHEF_LIEU, ID_SIG) VALUES
pluviom-# ('TN-11', 'Tunis', 'Tunis', 1),
pluviom-# ('TN-12', 'Ariana', 'Ariana', 2),
pluviom-# ('TN-13', 'Ben Arous', 'Ben Arous', 3),
pluviom-# ('TN-14', 'Manouba', 'Manouba', 4),
pluviom-# ('TN-21', 'Nabeul', 'Nabeul', 5),
pluviom-# ('TN-22', 'Zaghouan', 'Zaghouan', 6),
pluviom-# ('TN-23', 'Bizerte', 'Bizerte', 7),
pluviom-# ('TN-31', 'Béja', 'Béja', 8),
pluviom-# ('TN-32', 'Jendouba', 'Jendouba', 9),
pluviom-# ('TN-33', 'Le Kef', 'Le Kef', 10),
pluviom-# ('TN-41', 'Siliana', 'Siliana', 11),
pluviom-# ('TN-42', 'Kairouan', 'Kairouan', 12),
pluviom-# ('TN-43', 'Kasserine', 'Kasserine', 13),
pluviom-# ('TN-51', 'Sousse', 'Sousse', 14),
pluviom-# ('TN-52', 'Monastir', 'Monastir', 15),
pluviom-# ('TN-53', 'Mahdia', 'Mahdia', 16),
pluviom-# ('TN-61', 'Sfax', 'Sfax', 17),
pluviom-# ('TN-71', 'Kébili', 'Kébili', 18),
pluviom-# ('TN-72', 'Gabès', 'Gabès', 19),
pluviom-# ('TN-73', 'Médenine', 'Médenine', 20),
pluviom-# ('TN-81', 'Tataouine', 'Tataouine', 21),
pluviom-# ('TN-82', 'Gafsa', 'Gafsa', 22),
pluviom-# ('TN-83', 'Tozeur', 'Tozeur', 23),
pluviom-# ('TN-84', 'Sidi Bouzid', 'Sidi Bouzid', 24);
INSERT 0 24
pluviom=#

pluviom=# INSERT INTO region_naturelle (id, name) VALUES
pluviom-# (1, 'Nord Est'),
pluviom-# (2, 'Nord Ouest'),
pluviom-# (3, 'Centre Est'),
pluviom-# (4, 'Centre Ouest'),
pluviom-# (5, 'Sud Est'),
pluviom-# (6, 'Sud Ouest');

pluviom=# -- Nord Ouest - Jendouba
pluviom=# INSERT INTO publicf.station (name, id_gouvernorat, type_station, active, superficie_theissen) VALUES
pluviom-# ('JENDOUBA', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('BOU SALEM DRE', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('GARDIMAOU DRE', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('AIN DRAHAM SM', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('TABARKA FORET', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('BOU HEURTMA', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL),
pluviom-# ('Moyenne', (SELECT id FROM publicf.gouvernorat WHERE name_fr='Jendouba'), 'Pluviomètre', TRUE, NULL);
```

Figure 29 : An example of script to populate table

In the figure 28, the tables are inserted with actual data :

pluviom/postgres@PostgreSQL 16								
Query Query History Scratch Pad x								
1 SELECT * FROM publicf.GOUVERNORAT;								
Data Output Messages Notifications								
Showing rows: 1 to 48 Page No: 1 of 1								
	id [PK] integer	code character varying (50)	name_fr character varying (150)	chef_lieu character varying (150)	id_sig numeric (12,3)	id_region integer	superficie_km2 numeric (12,2)	
1	6	ARI	Ariana	Ariana	[null]	2	482.00	
2	7	BEN	Ben Arous	Ben Arous	[null]	2	761.00	
3	8	MAN	Manouba	Manouba	[null]	2	1137.00	
4	20	KEB	Kébili	Kébili	[null]	5	22102.79	
5	22	MED	Médenine	Médenine	[null]	6	8615.46	
6	1	JEND	Jendouba	Jendouba	[null]	1	3121.12	
7	2	BEJ	Beja	Beja	[null]	1	3697.23	
8	3	KEF	Le Kef	Le Kef	[null]	1	5109.27	
9	4	SIL	Siliana	Siliana	[null]	1	4588.87	
10	9	NAB	Nabeul	Nabeul	[null]	2	2783.75	
11	10	ZAG	Zaghuan	Zaghuan	[null]	2	2895.07	
12	11	BIZ	Bizerte	Bizerte	[null]	2	3591.66	
13	12	KAI	Kairouan	Kairouan	[null]	3	6650.29	
14	13	KAS	Kasserine	Kasserine	[null]	3	8161.42	
15	24	SDB	Sidi Bouzid	Sidi Bouzid	[null]	3	7372.26	
16	14	SOU	Sousse	Sousse	[null]	4	2624.35	
17	15	MON	Monastir	Monastir	[null]	4	994.37	
18	16	MAH	Mahdia	Mahdia	[null]	4	2809.93	
19	17	SFA	Sfax	Sfax	[null]	4	7001.05	

	id [PK] integer	code character varying (10)	name_en character varying (50)	name_fr character varying (50)	superficie_km2 numeric (12,2)	
1	1	NO	North West	Nord Ouest	16516.50	
2	2	NE	North East	Nord Est	11725.00	
3	3	CO	Centre West	Centre Ouest	22184.00	
4	4	CE	Centre East	Centre Est	13429.70	
5	5	SO	South West	Sud Ouest	35761.20	
6	6	SE	South East	Sud Est	55305.40	

	id [PK] bigint	code character varying (50)	name character varying (200)	latitude numeric (12,6)	longitude numeric (12,6)	altitude_m numeric (8,2)	type_station character varying (50)	active boolean	
1	1	1	JENDOUBA	[null]	[null]	[null]	Pluviomètre	true	
2	2	2	BOU SALEM DRE	[null]	[null]	[null]	Pluviomètre	true	
3	3	3	GARDIMAOU DRE	[null]	[null]	[null]	Pluviomètre	true	
4	4	4	AIN DRAHAM SM	[null]	[null]	[null]	Pluviomètre	true	
5	5	5	TABARKA FORET	[null]	[null]	[null]	Pluviomètre	true	
6	6	6	BOU HEURTMA	[null]	[null]	[null]	Pluviomètre	true	
7	7	7	BEJA INRAT	[null]	[null]	[null]	Pluviomètre	true	
8	8	8	MEJEZ EL BAB PF	[null]	[null]	[null]	Pluviomètre	true	
9	9	9	NEFZA DELEGATION	[null]	[null]	[null]	Pluviomètre	true	
10	10	10	BARRAGE SIDI SALEM	[null]	[null]	[null]	Pluviomètre	true	
11	11	11	TEBOURSOUK SM	[null]	[null]	[null]	Pluviomètre	true	
12	12	12	TIBAR SM	[null]	[null]	[null]	Pluviomètre	true	
13	13	13	KEF CMA	[null]	[null]	[null]	Pluviomètre	true	
14	14	14	OUED MELLEGUE K13	[null]	[null]	[null]	Pluviomètre	true	
15	15	15	SAKIET SIDI YOUSSEF	[null]	[null]	[null]	Pluviomètre	true	
16	16	16	KALAAT ESSENAM	[null]	[null]	[null]	Pluviomètre	true	
17	17	17	KSOUR ECOLE	[null]	[null]	[null]	Pluviomètre	true	
18	18	18	SERS AGRICOLE	[null]	[null]	[null]	Pluviomètre	true	

Figure 30 : Extract of the contents of the tables

After i inserted the average rainfall data into the corresponding tables using a **Python script** shown in the figure 29 . The Excel file containing the average daily rainfall was organized in a **matrix format**, where each row represents a station and each column corresponds to a day of the hydrological year represent 366 columns.

for exemple To import this data into the APPORT_MOYEN_STATION table in PostgreSQL, the format columns must be transformed into a **long format**, where each row represents a single station for a single day and at the end each station has one row per day.

```
import pandas as pd

df = pd.read_excel("C:/Users/KHEMIRA/Desktop/pfe/apport_stations.xls", header=0)

print(df.columns)

id_vars = ['id'] # colonne id
value_vars = [col for col in df.columns if col not in ['id', 'STATIONS']]

df_long = pd.melt(df, id_vars=id_vars, value_vars=value_vars,
                  var_name='jour_hyd', value_name='moy_apport')

df_long = df_long.rename(columns={'id': 'sta_id'})

df_long['jour_hyd'] = df_long['jour_hyd'].astype(int)
df_long['moy_apport'] = df_long['moy_apport'].astype(float)

print(df_long.head())

id_vars = ['id'] # ou ['id', 'STATIONS']
value_vars = [col for col in df.columns if col not in id_vars]

df_long = pd.melt(df, id_vars=id_vars, value_vars=value_vars,
                  var_name='jour_hyd', value_name='moy_apport')

df_long['jour_hyd'] = df_long['jour_hyd'].astype(int)
df_long['moy_apport'] = df_long['moy_apport'].astype(float)

print(df_long.head())

df_long = pd.melt(df, id_vars=['id', 'STATIONS'],
                  value_vars=[list_de_tous_les_jours],
                  var_name='jour_hyd', value_name='moy_apport')
df_long.to_csv("C:/Users/KHEMIRA/Desktop/pfe/apport_long.csv", index=False)
COPY apport_moyen_station (id, sta_id, date_ref, moy_apport)
FROM 'C:\Users\KHEMIRA\Desktop\pfe\apport_long.csv'
DELIMITER ',' CSV HEADER;
```

Figure 31 : Code used to import the average rainfall data of the stations from the Excel sheet into the Pluviom database

Table 8 : The number of data insred in the PLUVIOM database

Table noun	Number of data
station	128
gouvernorat	24
region_naturelle	6
apport_moyen_station	45 750
apport_moyen_gouvernorat	8784
apport_moyen_region_naturelle	2562

III. Description and Interpretation of Application Modules

III.1 System aarchitector

The development of the web platform for rainfall data management is based on a three-tier architecture ensuring the separation of the Frontend, the Backend, and the data layer. The image 33 shoven the architecture of the systeme

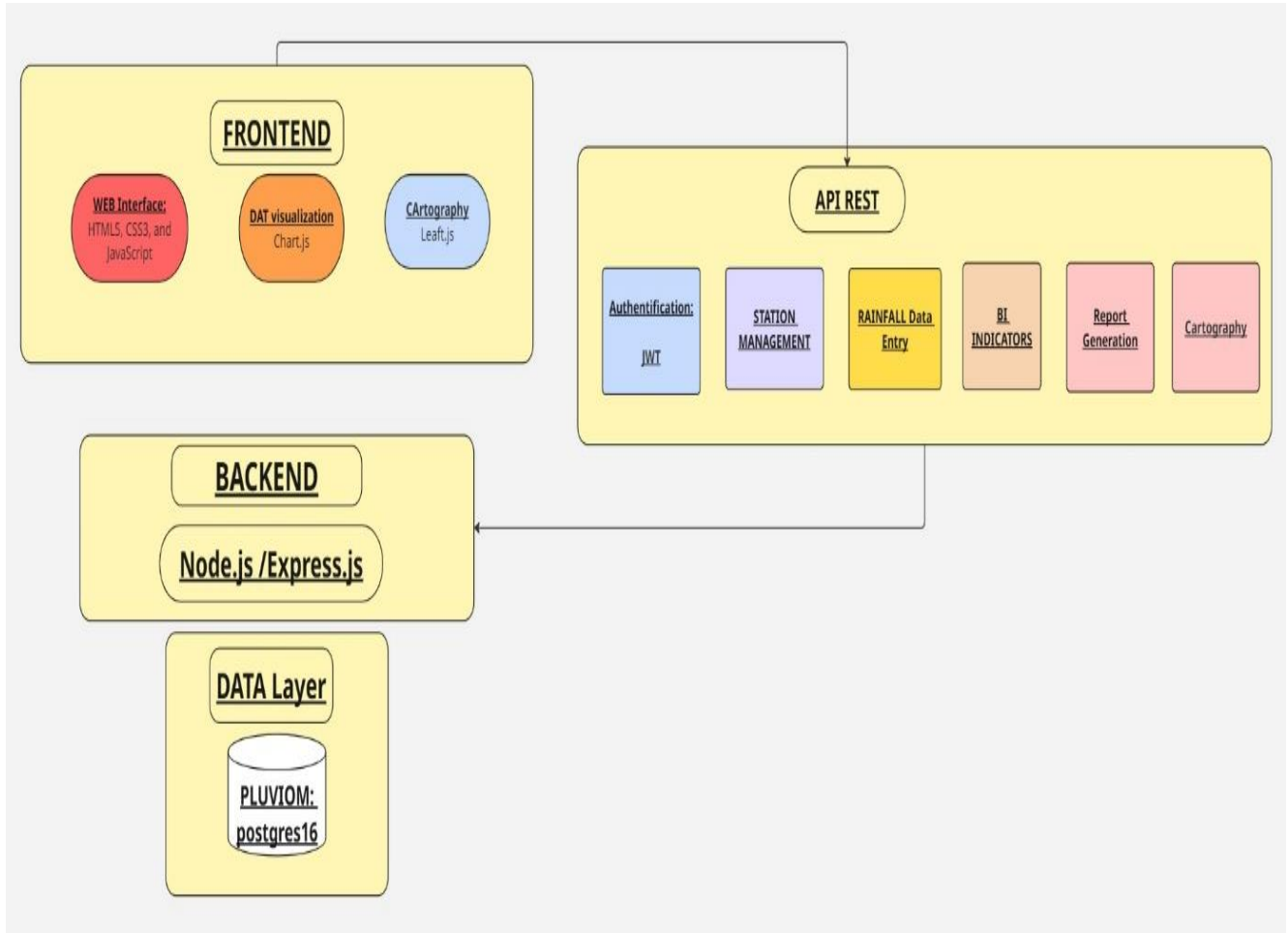


Figure 33: System Architector

III.1.1 Frontend:

The presentation layer is responsible for the user interface and interaction serves as the direct interface between the system and its users. It is designed to be interactive and dynamic, as requested by users during the diagnostic phase, to facilitate the consultation and entry of rainfall data by technicians and end users. The interface is developed with HTML5, CSS3, JavaScript and Bootstrap 5.

For data visualization Chart.js is used to represent pluviometric indicators and historical trends and Leaflet.js is used to manage and display station locations and rainfall maps interactively.

III.1.2 Backend :

The backend is the core of the application, managing business logic, calculations, and database access using Node.js with the Express.js

The frontend communicates with the backend via RESTful APIs for the real-time updates of DATA and information's PLUVOM data base, each route communicates with a corresponding table in the database to manage data operations.

The REST API is organized around the main entities of the system:

- ✓ Authentication
- ✓ Stations
- ✓ Rainfall Data
- ✓ BI Indicators
- ✓ Governorates and Natural Regions
- ✓ Import and export to generate reports
- ✓ Cartography

Each route communicates with a table in the database to manage data and provides a Key functionalities include:

- ✓ Secure authentication using JSON Web Tokens (JWT), with hierarchical access levels by governorate and central access.
- ✓ Management of stations and rainfall data with full CRUD operations (create, update, soft delete).
- ✓ Automatic generation of BI indicators, calculating the 14 columns of the rainfall bulletin based on historical data.
- ✓ This architecture ensures data consistency, service availability, and modularity, facilitating the future evolution of the application.
- ✓ Report generation using PDFKit, ExcelJS, and docx.
- ✓ Maps generation using Leaflet.js

III.2 Functional Flowchart

From the main window of the portal we can access the various integrated modules.

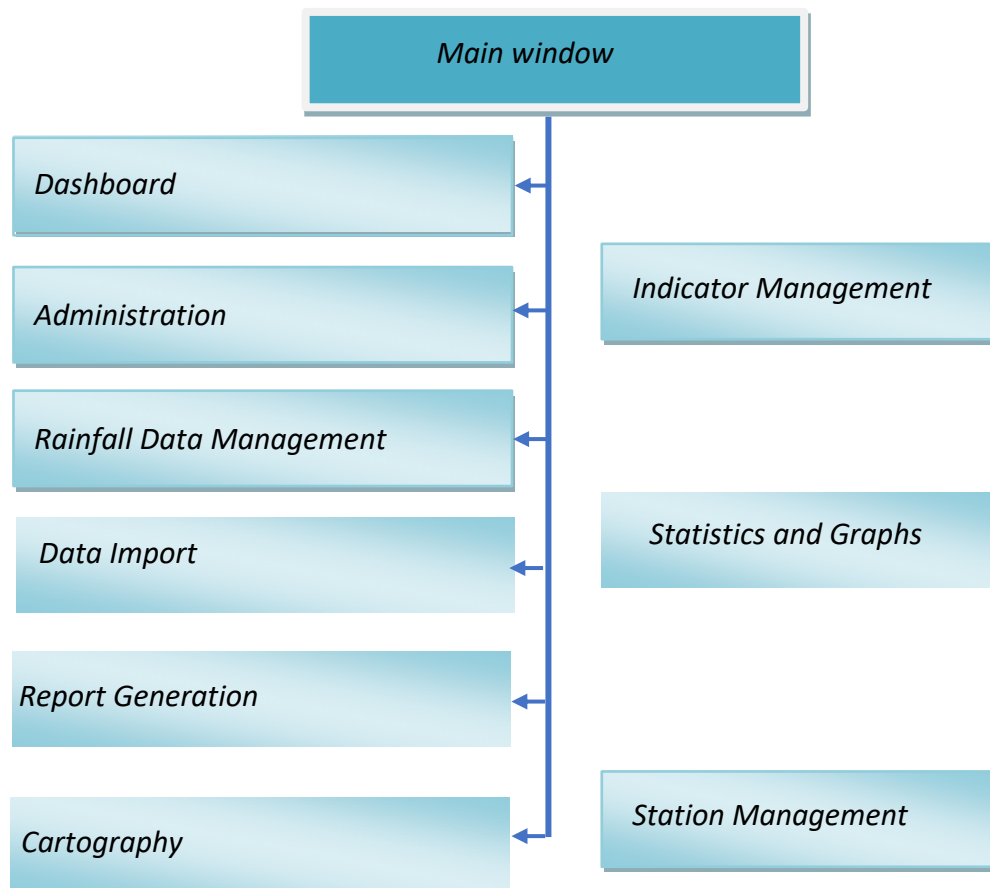


Figure 34: Functional Flowchart

III.3 The Application Modules

III.3.1 Authentication Module

The Authentication Module (Figure 20) allows access to the system with security and access control, ensuring that only authorized users can interact with the system and its various functionalities. It typically involves a login screen where users provide csername and password to access.

Figure 35:the Authentication Module

III.3.2 "Home Module"

The Home Module in figure 33 serves as the central dashboard and landing page for the Pluviometric Management Application. It provides the system's overview and the core functionalities available to the user. It contains also the key performance indicators (KPIs) as the number of station, The number of daily rainfall data entries during the current month, The calculated average cumulative rainfall and the last entry entry into the system.

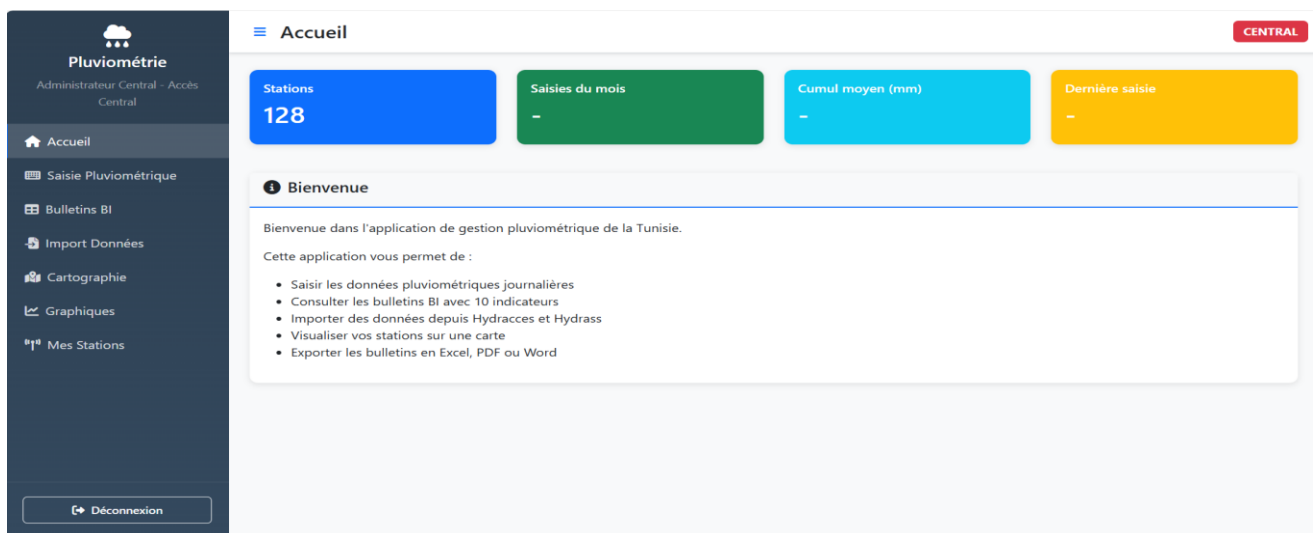


Figure 36: the Home Module

III.3.3 "Rainfall Data Entry" Module

In This module the users record the actual rainfall data collected from the stations and data can be shown in a table

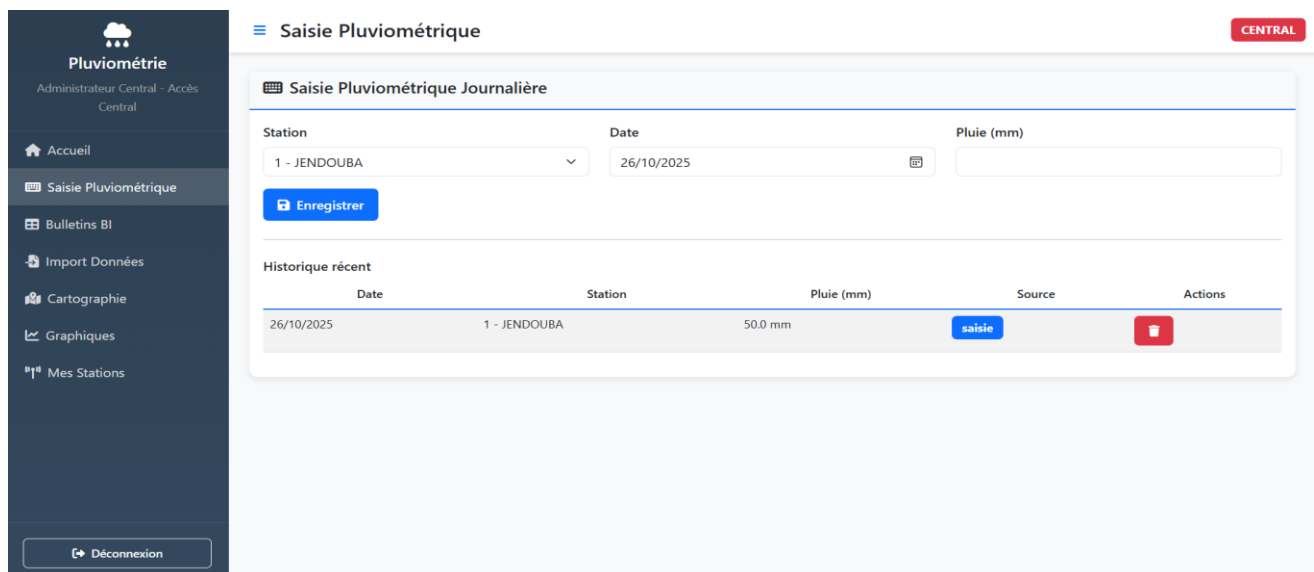


Figure 37: "Rainfall Data Entry" Module

III.3.4 "BI Report" Module

The BI module Generate all the indicators allowed by the rainfall bulletin based on pluviometric data entry

Type	Station	Date	Col 1 Pluie Jour	Col 2 Cumul Mois	Col 3 Moy. Mois	Col 4 %	Col 5 Cumul Période	Col 6 Moy. Période	Col 7 %	Col 8 Cumul N-1	Col 9 Diff.	Col 10 % Comp.
Station	1 - JENDOUBA	26/10/2025	50.0	50.0	50.0	100.0	50.0	50.0	100.0	0.0	50.0	0.0

Figure 38: "BI Report" Module

III.3.5 "IMPORT" Module

The import module dedicated to importing pluviometric data from hydracces and hydrass. And users can choose the source file to import CSV or XML format.

Import Hydracces

Importer des données pluviométriques depuis Hydracces

Choisir un fichier | Aucun fichier n'a été sélectionné

Importer

Import Hydrass

Importer des données pluviométriques (5 min) depuis Hydrass

Choisir un fichier | Aucun fichier n'a été sélectionné

Importer

Figure 39: "IMPORT" Module

III.3.6 Module «Cartographie »

The cartography module offers a spatial visualization of the stations and their location and we can visualize the daily data on an interactive map.

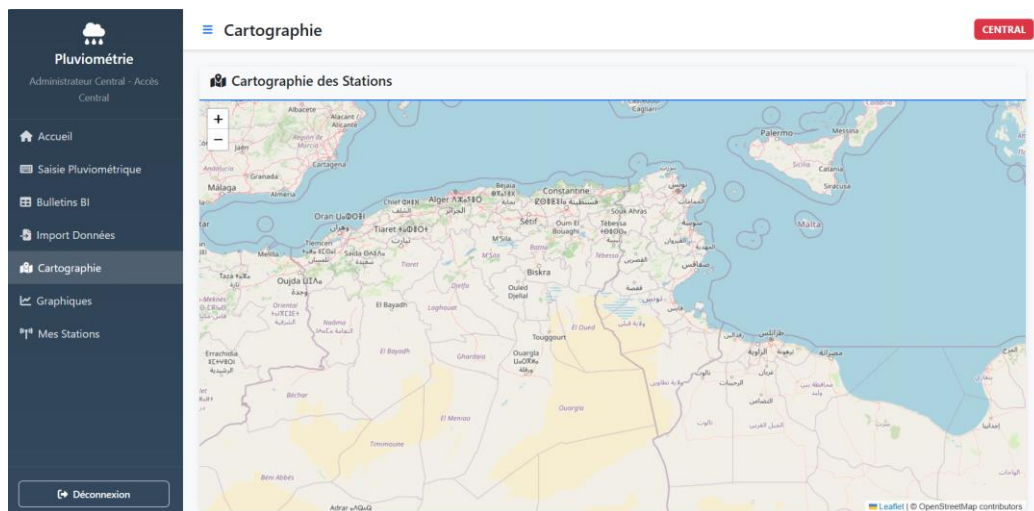


Figure 40:Module «Cartographie »

III.3.7 "Graphic" Module

The graphic module is the analytical visualization , it represent trends and comparisons of pluviometric data for deeper analysis

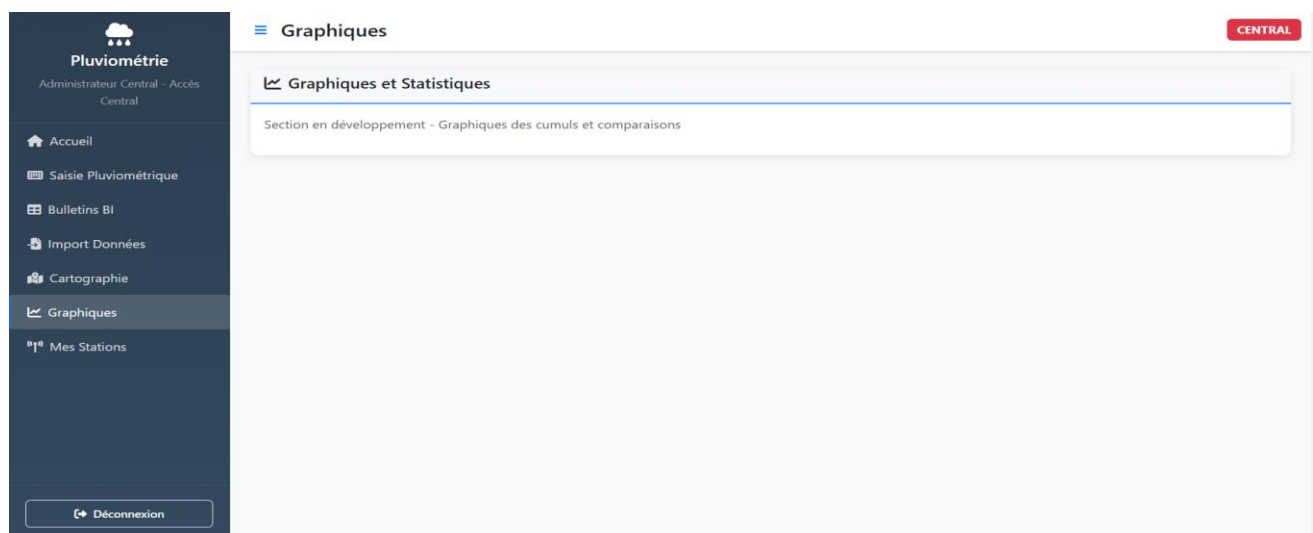


Figure 41: "Graphic" Module

III.3.7 "Station" Module

The station module is the management module for the station repository it manage the complete list of pluviometric stations and their attributes

Pluviométrie

Administrateur Central - Accès Central

Accueil

Saisie Pluviométrique

Bulletins BI

Import Données

Cartographie

Graphiques

Mes Stations

Déconnexion

Mes Stations

CENTRAL

Liste des Stations

Code	Nom	Latitude	Longitude	Altitude	Type	Statut
1	JENDOUBA	-	-	- m	Pluviomètre	Active
10	BARRAGE SIDI SALEM	-	-	- m	Pluviomètre	Active
100	EL GUETAR	-	-	- m	Pluviomètre	Active
101	ESSNED	-	-	- m	Pluviomètre	Active
102	METLAOUI SM	-	-	- m	Pluviomètre	Active
103	REDEYEF SM	-	-	- m	Pluviomètre	Active
104	SIDI AIECH	-	-	- m	Pluviomètre	Active
105	MOULARES SM	-	-	- m	Pluviomètre	Active
106	TOZEUR SM	-	-	- m	Pluviomètre	Active
107	TAMEGHZA,GN	-	-	- m	Pluviomètre	Active
108	DEGACHE MUNICIPALITE	-	-	- m	Pluviomètre	Active
109	NEFTA	-	-	- m	Pluviomètre	Active

Figure 42: "Station" Module

Conclusion générale

This project has laid the foundation of a web centralized system for rainfall data management in Tunisia. By replacing a manual and error-prone process with a web application built around a postgres database, we have demonstrated the feasibility and relevance of an integrated approach.

The current solution meets the immediate needs for data collection, centralization, and consultation, representing a significant step forward for the General Directorate for Water Resources (DGRE) and its partners.

However, this work is only the first step toward a deeper digital transformation. The future work is extensive and revolves around several strategic axes to evolve this tool into a national knowledge management system for water resources.

The next step will be to deploy the application in a controlled test environment with end-users (CRDA agents, ministry staff). This phase will allow for the collection of user feedback to identify bugs, optimize usability, and validate the relevance of the features. A continuous improvement cycle based on this feedback will be established to refine the application.

Once the application is stable, the objective will be to make it accessible to a wider audience. This will involve concrete technical steps:

- **Hosting on a physical server:** To ensure performance, security, and data sovereignty, the application will be hosted on a dedicated server within the ministry's infrastructure.
- **Purchasing a domain name** (www.pluviometrie.tn) will be acquired to facilitate access to the platform.
- The application will be officially launched and made accessible to the general public, including farmers, researchers, the media, and citizens interested in water data.

the long-term vision is to transcend simple data management to create a Knowledge Management System (KMS) dedicated to water resources. The **PLUVIOM** application will be the first module of this integrated platform. Future upgrades will aim to integrate other types of hydrological data:

- Groundwater table levels.

- River flow rates.
- Dam filling rates.
- Water quality.

This KMS will not only centralize information but also contextualize, analyze, and transform it into actionable knowledge. It will integrate predictive analysis tools, dynamic dashboards, and simulation models to support strategic decision-making in water management, thereby making a significant contribution to Tunisia's water security

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Annex 1

SECADENORD :

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	Weekly and after significant rainfall events
	For what specific purposes do you use the rainfall data?	Evaluating catchment responses and runoff
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	Generally sufficient
	Do you face any difficulties accessing or comparing historical data?	no built-in option to compare current data with historical data beyond the previous year.
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	Acceptable and reliable overall
	Are you satisfied with the timeliness of the bulletin publication?	Generally timely
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	web-based interactive platform
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	Advanced tools for trend and anomaly detection, including curves and charts to better visualize the data
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	minor inconsistencies
	What improvements could make data consultation and analysis easier for you?	Interactive dashboards with filtering by station, region, and period

NAME
Function

Sous-Directeur
Etudes et Programmation Stratégique
Chaker BEN ZID



Water Recharge departement

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	I consult the rainfall bulletin once a day.
	For what specific purposes do you use the rainfall data?	I use the rainfall data to plan my daily activities, monitor weather conditions
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	yes
	Do you face any difficulties accessing or comparing historical data?	yes
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	I would evaluate the quality of the current rainfall data as generally good it is fairly accurate, reliable, and easy to read, although occasional updates or delays can occur.
	Are you satisfied with the timeliness of the bulletin publication?	yes
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	web interface, mobile app
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	Designing an interactive and dynamic platform for accessing information

Sous-directeur
de la Recherche Artificielle
Foudhaili Hedia

NAME : Ben Slim Khaoula

FUNCTION : Department of monitoring and control of hydraulic boreholes

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	Occasionally
	For what specific purposes do you use the rainfall data?	To schedule a trip/a field visit
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	Basic/the minimum necessary
	Do you face any difficulties accessing or comparing historical data?	Difficulties for comparing historical data and accessing at any time
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	The readability is limited because it is static and too information-heavy
	Are you satisfied with the timeliness of the bulletin publication?	no
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	Web interface
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	Interactive maps showing station locations and rainfall distribution
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	not tested

NAME : Ben Slim Khaoula

FUNCTION :Department of monitoring and control of hydraulic boreholes

	What improvements could make data consultation and analysis easier for you?	-Effective rainfall the precipitation that actually contributes to recharging groundwater aquifers. -Interactive maps showing station locations and rainfall distribution in real time -Time series charts and statistical analyses -Short-term forecasts
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رئيس مصلحة مراقبة ومتابعة
التنقيب عن المياه
حمولة بن سليم احمد منصوري

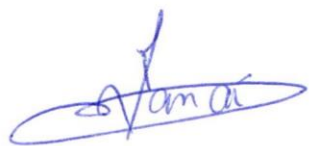
NAME *FRIJI Narene*

Function *Responsible of rainfall report "monthly."*

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	Every day
	For what specific purposes do you use the rainfall data?	Study the rainfall over a period of days or months
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	yes
	Do you face any difficulties accessing or comparing historical data?	As long as I am under the supervision of the studies and research department, I want all the data to be accessible to me.
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	Data is improving significantly
	Are you satisfied with the timeliness of the bulletin publication?	yes
	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	web interface, mobile app
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	Access and enhancement of data
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	Previously I noticed some errors in the Rain bulletin, but they have been corrected. This has made me cautious when checking the Rain bulletin.
	What improvements could make data consultation and analysis easier for you?	Monthly and annual rainfall averages must be updated.

Name: Ammar Manai

Fonction: Technicien



rainfall directory responsible

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	I check the rainfall forecast daily, especially during periods of deteriorating weather conditions.
	For what specific purposes do you use the rainfall data?	Rainfall data is used to take necessary precautions before floods occur in the event of heavy rainfall over a short period.
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	fairly sufficient
	Do you face any difficulties accessing or comparing historical data?	There are some difficulties, but they can be overcome.
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	I evaluate the quality of the current data (accuracy, reliability, and readability).
	Are you satisfied with the timeliness of the bulletin publication?	no
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	Web interface
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	What new features or improvements would you like to see in a future system?
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	no
	What improvements could make data consultation and analysis easier for you?	Provide a graph or map of rainfall distribution to facilitate data analysis.

NAME Abir BELLALI



Function Doctor-engineer / Chief of Service

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	daily
	For what specific purposes do you use the rainfall data?	Assessment of surface and groundwater resources
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	yes
	Do you face any difficulties accessing or comparing historical data?	No, I didn't face any difficulties collecting historical rainfall data.
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	Mean values of the dataset
	Are you satisfied with the timeliness of the bulletin publication?	yes
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	web interface
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	a monthly rainfall variation will be calculated at the end of each month based on daily rainfall data
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	No errors in the bulletin
	What improvements could make data consultation and analysis easier for you?	to develop a platform that enables easier access to data

NAME : Houda Rzigui



Function: Principal Engineer (GDRE)

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	5 to 7 times a year.
	For what specific purposes do you use the rainfall data?	Net Recharge estimation
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	No; it needs improvement.
	Do you face any difficulties accessing or comparing historical data?	Yes, The table is difficult to interpret because the column titles are not very clear. Also, comparing historical data from a selected station is hard. better to use access!
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	Reliability and Accuracy are acceptable! Readability is not satisfactory enough!
	Are you satisfied with the timeliness of the bulletin publication?	Yes; it is fine.
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	Application or web interface
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	- System must allow easier data export (select station-period-maximums.....). - improve data organization and visualization: (automatic charts, summary statistics,.....)
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	Yes. These errors made it difficult to calculate recharge amounts. it needs extra time and effort checking and correcting the data before using it.
	What improvements could make data consultation and analysis easier for you?	- More organized data structure, - clear labeling of variables - automatic summaries or graphs. - possibility of selecting station or time period to view specific rainfall trends and detect gaps on data.

NAME JILANI dhifli



Function (REAL-time monitoring responsible)

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	Daily
	For what specific purposes do you use the rainfall data?	I use it to validate the hydrass data and compare each value's station in hydrass with data's buletin
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	Yes I sufficient
	Do you face any difficulties accessing or comparing historical data?	Yes I need to consult my email and made a reserch to a specific day to access
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	reliability
	Are you satisfied with the timeliness of the bulletin publication?	no
Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	web interface or mobile app
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	web System
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	Yes, you must correct the situation and th work and it is not good for the administration's image
	What improvements could make data consultation and analysis easier for you?	Graphic (compare monthly data and annuel data

Theme	Question	Answer / Notes
Usage and Frequency	How often do you consult the rainfall bulletin?	Twice per year
	For what specific purposes do you use the rainfall data?	identify the months with significant rainfall in order to correlate these data with fluctuations in the static level of deep groundwater, and to approximately determine the volume of precipitation that reaches both shallow and deep aquifers
Data Quality and Accessibility	Is the current data sufficient for your operational needs?	yes
	Do you face any difficulties accessing or comparing historical data?	No
	How would you evaluate the quality of the current data (accuracy, reliability, readability)?	The rainfall bulletin contains a lot of unnecessary data that a layperson or a casual visitor would find difficult to understand The rainfall bulletin is full of padding However, the data presented in this bulletin is very accurate and meticulous
	Are you satisfied with the timeliness of the bulletin publication?	yes

Formats and Access	Which format or platform do you prefer for accessing rainfall information (e.g., PDF, web interface, mobile app)?	Mobile app
Suggestions and Improvements	What new features or improvements would you like to see in a future system?	In the future, I would like the bulletin to show the location of the rain gauges, their corresponding rainfall data, and the nearest monitoring well in the region of interest aiming to quantify the impact of precipitations in both shallow and deep groundwater.
	Have you ever noticed inconsistencies or errors in the bulletin? If yes, how did they affect your work or decisions?	Other than the typos, it is possible to find rain gauges that have no recorded data
	What improvements could make data consultation and analysis easier for you?	- <i>Simplified Tables</i>: The core bulletin could have a summary table with only the most critical data; - An Interactive Map: This is the most critical improvement. Instead of tables of numbers, I want to click on a map of the region.

Dalila JELLALIA

INGENEER IN CHIEF

2025	لغز
صباح	7

الوضعية الوظيفية	للأطوار	الى غاية
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طار بعض المحطات الفرجية: مد كساب بأجرة 28,8م سيدي بويكر زهوان 28م القلعة الكبيرة سوسة 11م
 (الذين المشي 7م جيلان العسرين 36م القليلة توار هم

[illegible]

2000

المحطة	العدد	النسبة (%)	البيانات									
			البيانات					البيانات				
			1	2	3	النسبة (%)	النسبة (%)	4	5	6	النسبة (%)	النسبة (%)
البيانات	العدد	النسبة (%)	1	2	3	النسبة (%)	النسبة (%)	4	5	6	النسبة (%)	النسبة (%)
100°(12/5)=13	12	100%	100°(9/5)=11	10	100%	100°(6/5)=7	100%	100°(3/2)=4	3	100%	100°(1/2)=1	100%
100%	301	204%	11.3	111.7	14.7	141%	224.6	317.7	60%	32.8	20.0	1.0
100%	240	200%	10.0	127.0	22.0	142%	183.7	261.5	105%	30.4	32.0	0.0
100%	285	339%	28.0	80.0	8.5	143%	190.2	271.3	118%	26.3	33.5	1.0
95%	214	337%	-3.5	82.0	16.0	120%	174.7	209.0	83%	23.8	12.5	0.0
157%	200	331%	29.0	96.5	16.0	198%	196.9	326.2	278%	17.0	47.0	0.0
59%	207	153%	0.0	92.0	9.0	82%	172.1	140.3	36%	25.0	9.0	0.0
95%	193	362%	-11.0	42.5	11.0	102%	164.0	190.5	0%	30.5	0.0	0.0
162%	236	279%	8.7	87.7	14.2	133%	182.3	241.8	90%	25.4	22.9	0.9
97%	354	209%	1.0	165.0	6.0	112%	306.4	344.5	21%	33.5	7.0	2.5
111%	370	206%	0.5	196.5	4.0	129%	315.7	405.8	14%	33.3	4.5	4.5
96%	330	147%	9.0	219.5	5.0	111%	290.7	322.5	42%	33.4	14.0	2.0
114%	330	243%	2.0	159.0	7.0	135%	286.2	396.0	30%	30.1	9.0	6.0
95%	303	230%	4.5	125.0	2.0	117%	253.7	294.0	21%	30.8	6.5	4.0
103%	340	203%	3.4	173.4	4.8	121%	296.5	362.0	25%	32.2	8.2	3.8
151%	330	172%	-6.0	296.0	10.0	171%	346.3	510.0	16%	30.6	5.0	3.0
152%	369	230%	-7.0	236.0	6.0	170%	339.0	591.0	3%	31.0	1.0	0.0
146%	371	230%	-7.0	225.0	7.0	160%	334.8	536.0	0%	32.4	0.0	0.0
134%	369	205%	0.0	235.0	8.0	150%	320.8	481.0	25%	32.4	8.0	3.0
145%	369	219%	-4.8	248.5	9.5	163%	320.7	622.0	11%	31.6	3.5	1.6
136%	343	174%	-1.1	271.6	6.1	153%	308.6	472.9	17%	28.8	5.0	0.0
167%	302	187%	-4.0	269.0	6.0	181%	278.1	504.0	9%	22.7	2.0	0.0
132%	248	194%	19.5	167.8	4.5	155%	210.9	326.0	88%	27.4	24.0	1.0
129%	292	194%	9.0	194.6	5.0	150%	251.7	376.7	45%	30.8	14.0	0.0
142%	296	198%	6.9	226.8	6.4	160%	262.3	419.9	41%	27.4	11.3	0.3
110%	231	310%	-15.0	96.0	16.0	130%	201.3	273.3	4%	25.0	1.0	0.0
151%	204	414%	-12.5	74.0	10.0	170%	174.2	306.4	26%	21.2	5.5	0.0
151%	199	500%	13.0	60.0	17.0	170%	166.4	300.0	137%	21.9	30.0	0.0
133%	177	363%	-19.0	68.0	22.0	152%	155.2	235.0	10%	16.7	3.0	0.0
120%	237	230%	-16.0	121.0	16.0	130%	207.2	295.5	0%	25.0	0.0	0.0
130%	208	291%	-5.5	96.5	24.0	153%	198.0	307.0	69%	20.8	18.5	0.0
175%	219	250%	-3.5	153.5	14.0	209%	153.4	393.5	51%	22.4	11.5	1.0
140%	211	315%	-6.2	94.0	16.1	162%	182.5	295.9	46%	21.9	9.9	0.1
62%	170	161%	-6.0	85.5	14.5	75%	136.2	105.5	33%	16.9	6.5	1.0
85%	142	319%	-6.5	38.5	5.5	102%	120.2	123.0	28%	17.8	5.0	0.0
100%	204	291%	6.5	76.0	21.0	129%	170.9	221.0	106%	26.0	27.5	0.0
85%	114	223%	-7.0	43.5	12.0	103%	94.4	97.0	38%	13.3	5.0	0.0
91%	130	200%	-6.0	60.5	15.0	118%	109.1	126.0	65%	15.5	10.0	0.0
107%	157	303%	3.5	55.5	11.5	133%	126.3	160.0	80%	18.9	15.0	0.0
71%	227	300%	8.0	42.5	7.0	94%	171.3	161.3	67%	22.4	15.0	2.0
87%	166	242%	-4.4	54.6	12.4	108%	132.9	143.1	63%	18.1	12.0	0.4
110%	94	235%	-19.7	48.2	22.0	142%	76.5	109.6	23%	9.8	2.3	0.0
80%	118	275%	-3.0	33.5	10.5	101%	82.2	83.5	50%	15.0	7.5	2.0
114%	97	265%	-20.0	43.5	22.0	139%	80.1	111.0	19%	10.5	2.0	0.0
98%	95	302%	-15.5	31.0	15.5	121%	77.4	93.5	0%	10.6	0.0	0.0
101%	101	284%	-14.6	38.6	17.6	125%	81.6	101.7	26%	11.6	3.0	0.5
81%	101	174%	-20.0	47.5	20.0	94%	86.1	82.5	0%	15.3	0.0	0.0
101%	74	163%	-17.0	46.2	17.0	117%	83.9	75.1	0%	8.1	0.0	0.0
62%	88	107%	-19.2	56.9	19.2	73%	77.8	80.7	0%	12.3	0.0	0.0
83%	88	148%	-16.7	50.2	16.7	86%	76.8	72.8	0%	11.9	0.0	0.0
65%	215	133%	-66.5	137.5	56.5	97%	169.4	103.5	0%	21.1	0.0	0.0
69%	140	140%	-22.0	68.5	22.0	78%	130.1	101.5	0%	17.2	0.0	0.0
125%	164	253%	-22.0	90.5	23.0	139%	164.6	229.0	5%	19.3	1.0	0.0
111%	200	141%	-29.0	156.5	29.0	121%	162.4	221.0	0%	31.7	0.0	0.0
89%	166	162%	-32.4	113.3	32.6	110%	166.6	193.6	1%	22.3	9.3	0.0
63%	164	63%	-57.2	217.3	57.2	90%	152.8	137.2	0%	24.9	0.0	0.0
47%	225	82%	-55.3	126.0	55.3	51%	207.3	105.5	0%	35.6	0.0	0.0
79%	168	104%	-15.0	142.1	15.0	94%	175.8	148.3	0%	24.1	0.0	0.0
47%	239	77%	-22.0	145.8	22.0	50%	225.9	112.9	0%	22.8	0.0	0.0
74%	223	102%	-56.0	161.8	56.0	79%	210.6	164.5	0%	18.2	0.0	0.0
84%	268	84%	-41.1	168.0	41.1	69%	184.4	133.7	0%	25.1	8.0	0.0
100%	127	141%	-26.7	96.2	26.7	121%	113.8	130.0	0%	23.9	0.0	0.0
67%	131	102%	-31.2	86.0	31.2	74%	119.6	86.1	0%	24.4	0.0	0.0
150%	83	477%	-1.8	27.1	1.8	179%	72.3	129.4	0%	13.6	0.0	0.0
100%	113	168%	-30.8	73.4	30.8	110%	101.9	118.6	0%	20.6	0.0	0.0

محطة تابعة للحمود الوطني ترصد الجوي *

المقارنة بالمحطات و المدن الفلاحية المتعددة حسب الجهات الطبيعية

المحطة	العدد	النسبة (%)	البيانات									
			البيانات					البيانات				
			1	2	3	النسبة (%)	النسبة (%)	4	5	6	النسبة (%)	النسبة (%)
البيانات	العدد	النسبة (%)	1	2	3	النسبة (%)	النسبة (%)	4	5	6	النسبة (%)	النسبة (%)
100°(12/5)=13	12	100%	100°(9/5)=11	10	100%	100°(6/5)=7	100%	100°(3/2)=4	3	100%	100°(1/2)=1	100%
النسبة (%)	301	204%	11.3	111.7	14.7	141%	224.6	317.7	60%	32.8	20.0	1.0
79%	530.9	140%	29.5	300.5	6.4	96%	430.8	421.9	69%	51.6	35.9	23.0
101%	504.4	191%	16.7	296.3	2.9	110%	430.9	506.2	48%	41.2	19.7	13.6
77%	285.4	215%	16.9	102.1	9.0	104%	210.5	219.9	92%	28.0	25.8	8.4
132%	268.1	235%	-3.7	148.5	12.1	152%	230.8	348.6	36%	25.8	9.4	1.0
90%	103.0	100%	-14.2	45.1	17.2	104%	89.2	82.4	23%	13.3	3.0	0.2
94%	137.9	144%	-25.4	90.0	25.4	104%	125.0	129.7	0%	21.6	0.0	0.0
92%	231.6	175%	-6.7	123.1	16.3	109%	194.6	212.9	41%	25.6	10.8	4.8